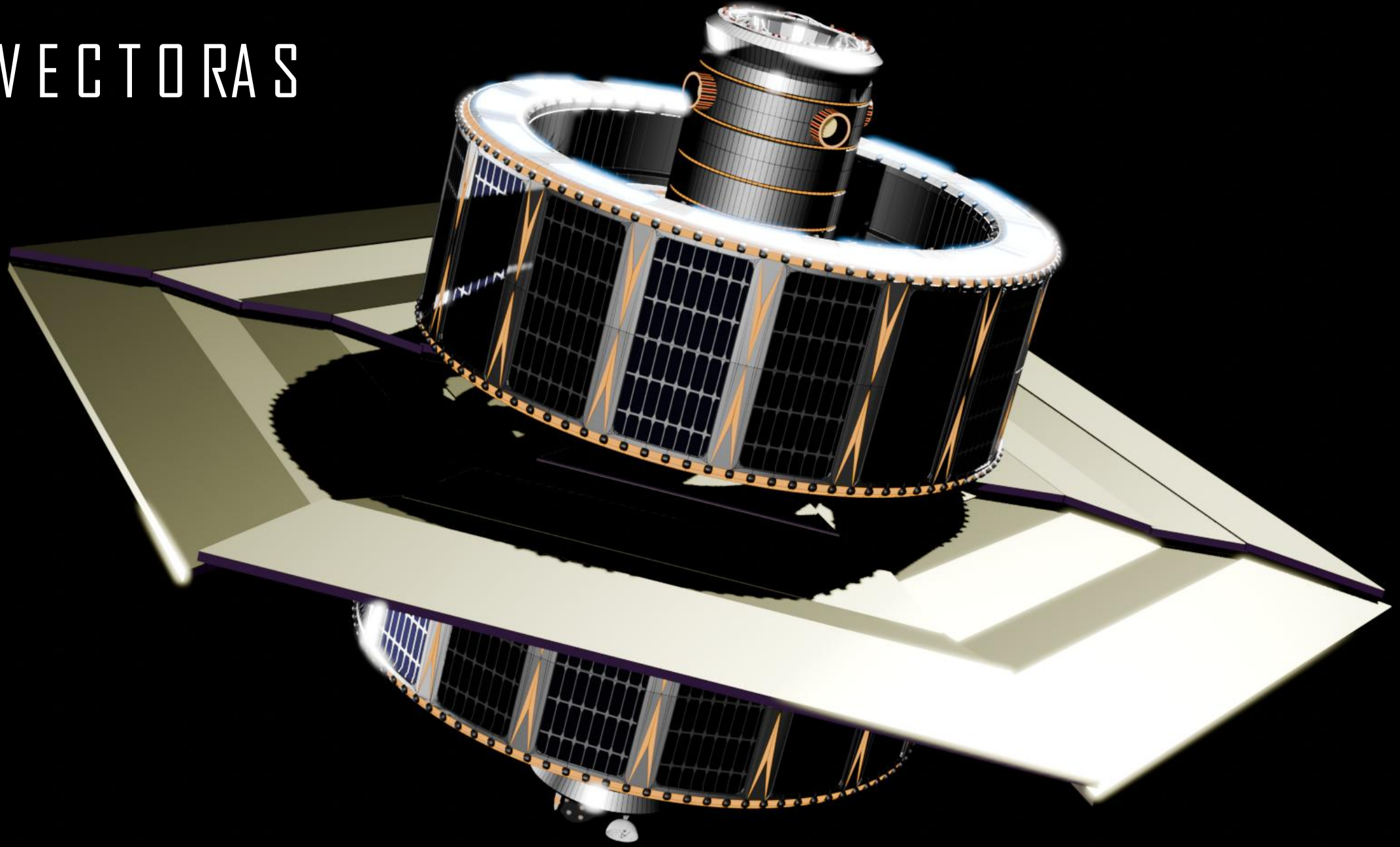
BENEVECTORAS

VULTURE AVIATION



CONSTRUCTION LOCATION

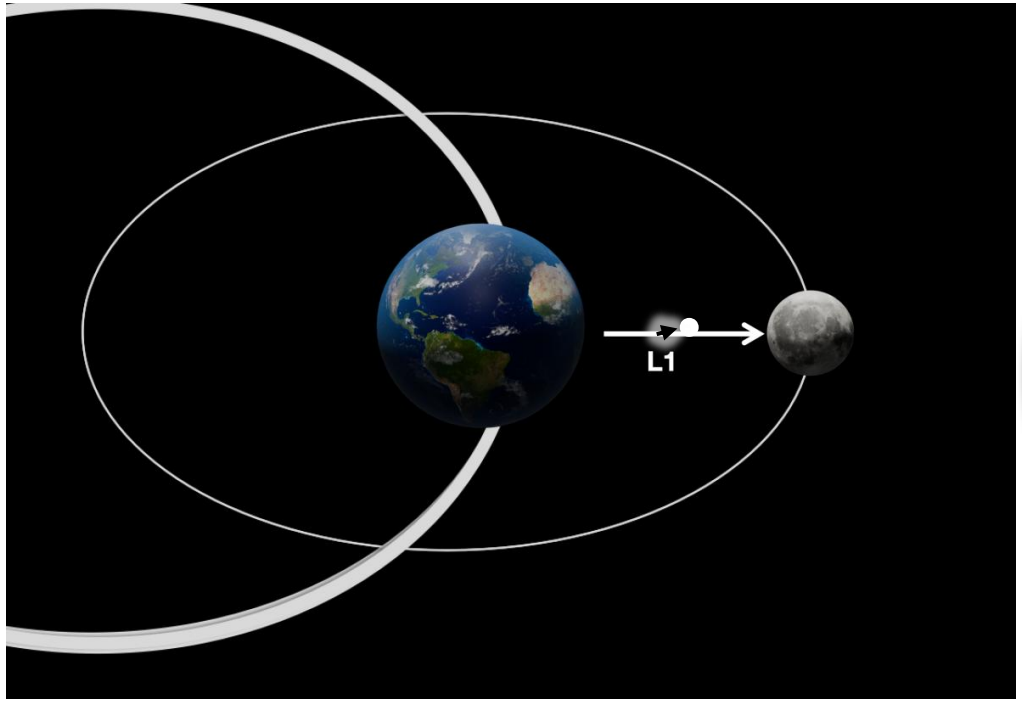


Fig : Earth-Moon L1 Point, Source: Kelly Z. Bilal Q., Blender

- Stationkeeping will be done at the Earth-Moon L1 point.
- Halo orbits used to minimize energy requirements.
- Small adjustments done with ion thrusters to counter gravitational disturbances and ensure settlement remains stable in orbit during construction.

Resource Access
Close to Earth and lunar materials.

Microgravity
Easier assembly of large structure

Transition Efficiency :
Simplifies movement to operational orbit.

Low Maintenance
Minimal energy for stationkeeping.

OPERATIONAL PLAN

Initial Orbital Insertion (into the Aldrin Cycler):
Using propulsion systems and gravity-assist maneuvers near Earth and Mars.

Stationkeeping:
Ion thrusters used to make periodic trajectory adjustments, ensuring precise alignment with the cycler path.

Space Tug Operations: Fleet of space tugs based at Benevectoras will transfer settlers/cargo during flybys and manage orbital corrections when required.

EXTERNAL CONFIGURATION

The RFP requests proposals for designing, developing, and operating the "Benevectoras" space settlement in cycler orbit between Earth and Mars. The settlement will use solar sails for propulsion, with space tugs assisting near Earth and Mars to optimize transportation efficiency.

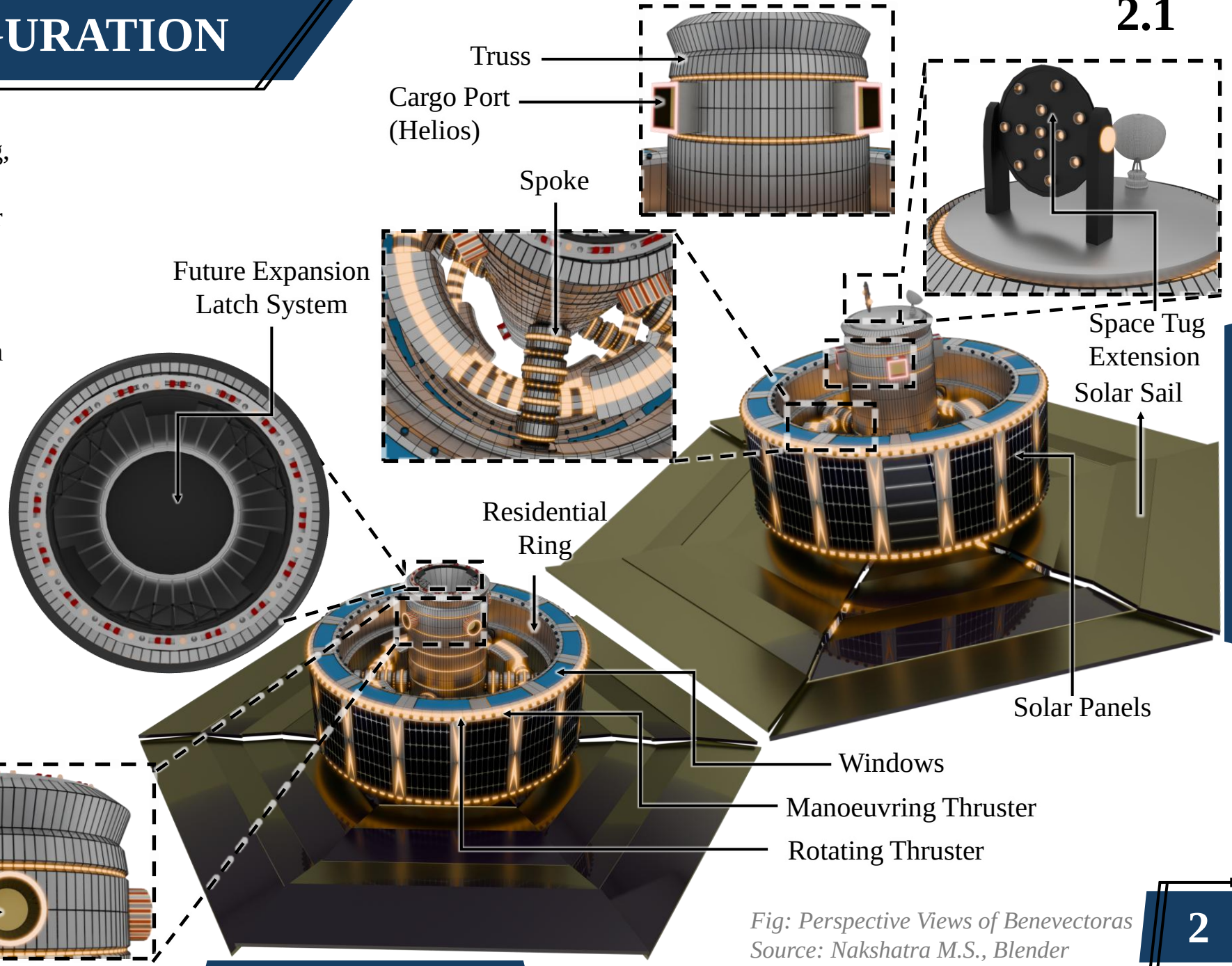


Fig: Perspective Views of Benevectoras
Source: Nakshatra M.S., Blender

EXTERNAL CONFIGURATION

2.1,4.0

- The structure consists of a non-rotating axle and two rotating rings connected to it through spokes.
- On the non-rotating axle, six fully deployable solar sails with a specialized retracting system, are arranged in a circle around the outer circumference.
- A latch system for future expansions is present. Space tugs storage and the circular space tug interface is present at the front end of the settlement.

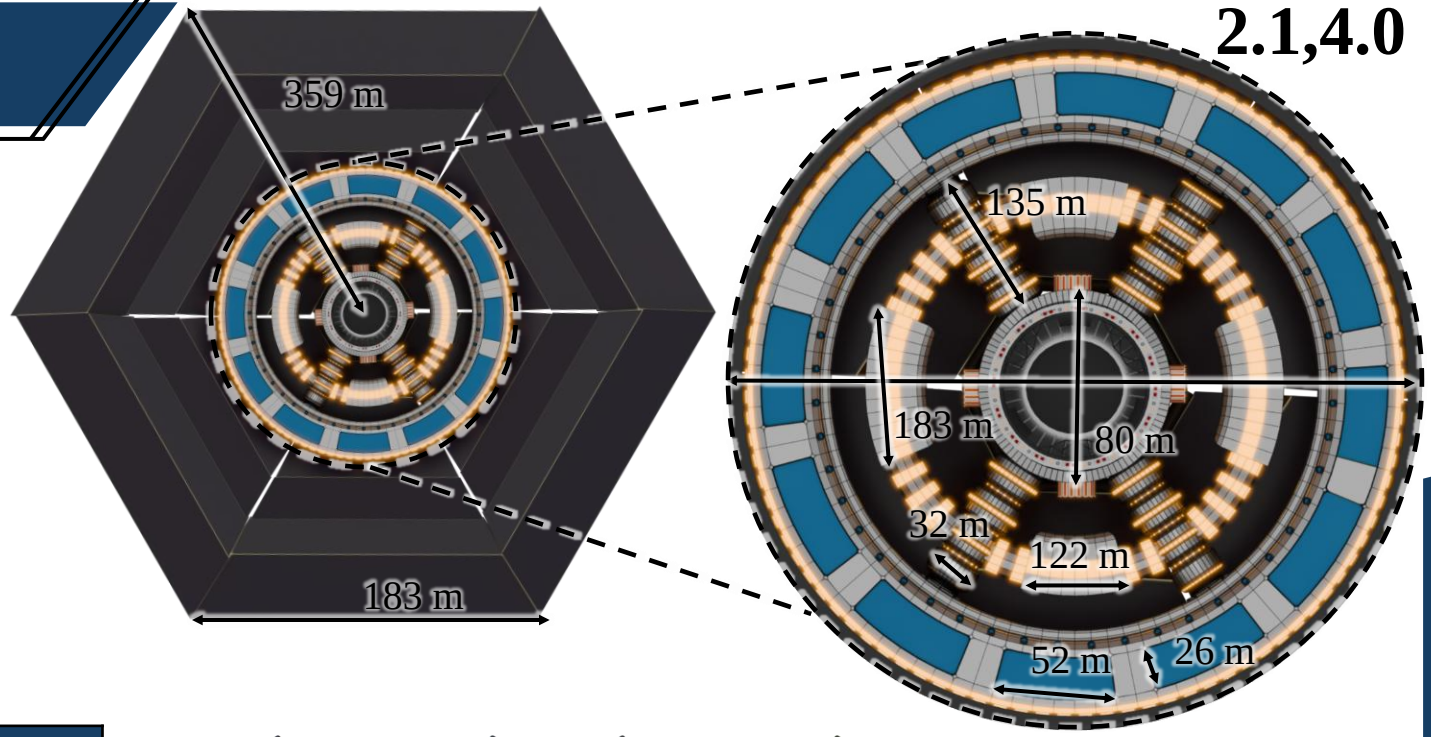


Fig: Front View Of Benevectoras
Source: Nakshatra M.S, Blender

Structural Components	Uses	Volume (m^3)
Rings	Consist of Residential and commercial areas.	17,761,308
Axle	Contains space tug docks, observatory, cargo and passenger docks	2,714,336
Spokes	Connects rings to axle and contains rotational interface	868,584

Fig: Major Structural Components
Source: Sumay M., Microsoft PPT1

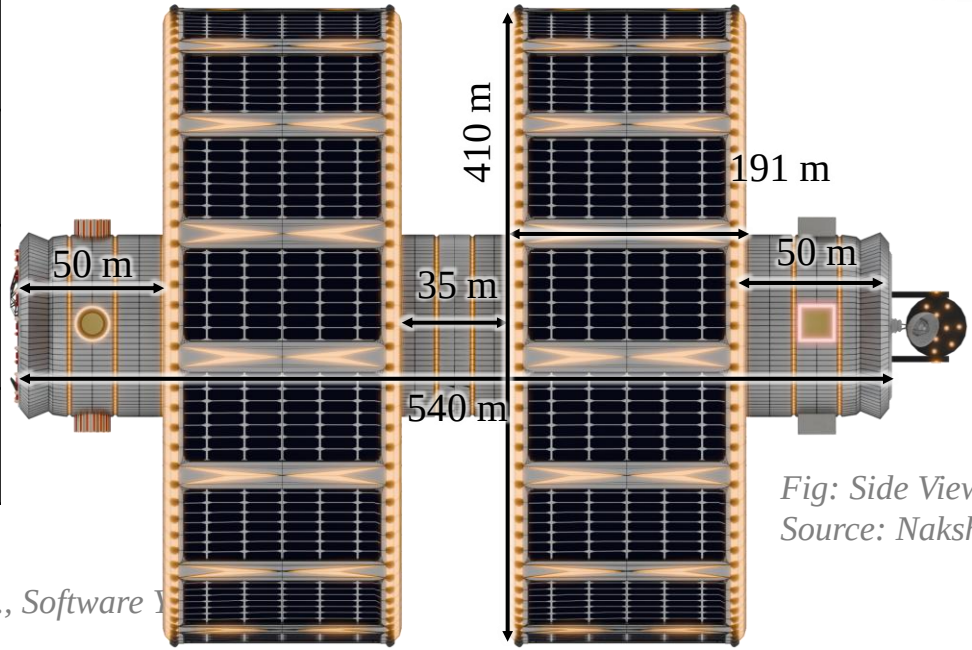


Fig: Side View Of Benevectoras
Source: Nakshatra M.S, Blender

Fig: X
Source: Ali Abbas R., Software Y

SOLAR SAIL SYSTEM

- 6 Solar Sails, connected to the axle in between the 2 rotating rings will propel the settlement.
- They’re made up of lightweight materials and are held up by multiple struts on each side.
- All solar sails are retractable and align parallelly to the non-rotating axle.
- Each strut is connected to a larger retractable rod, which can rotate entire individual solar sails for directional mobility and for stowage.

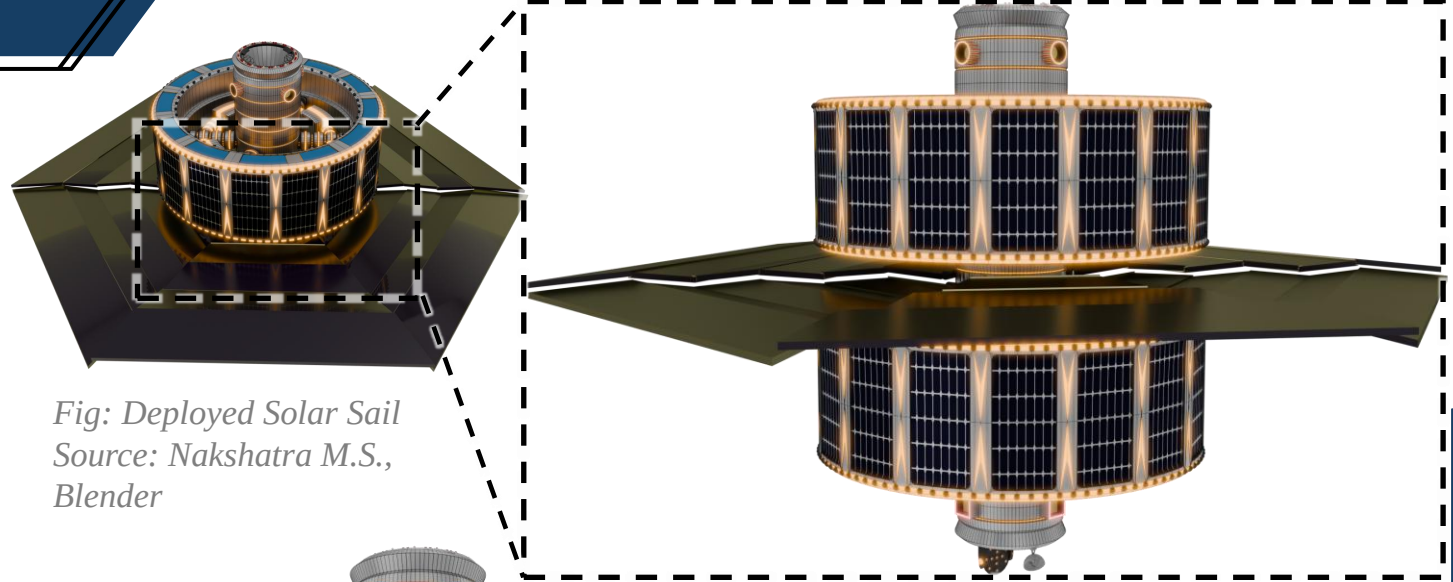


Fig: Deployed Solar Sail
Source: Nakshatra M.S., Blender

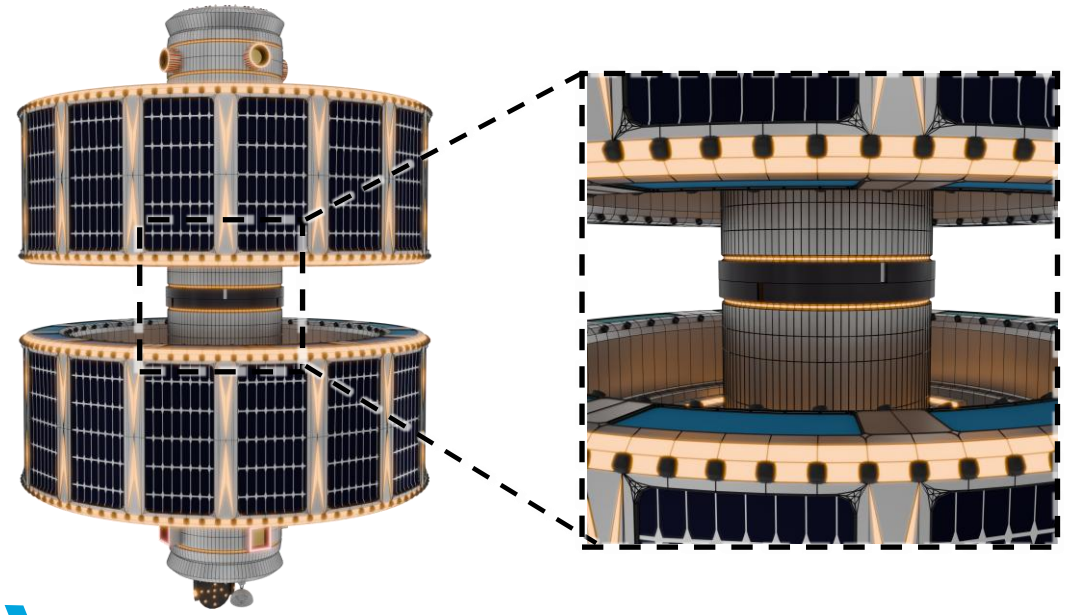


Fig: Stowed Solar Sail
Source: Nakshatra M.S., Blender

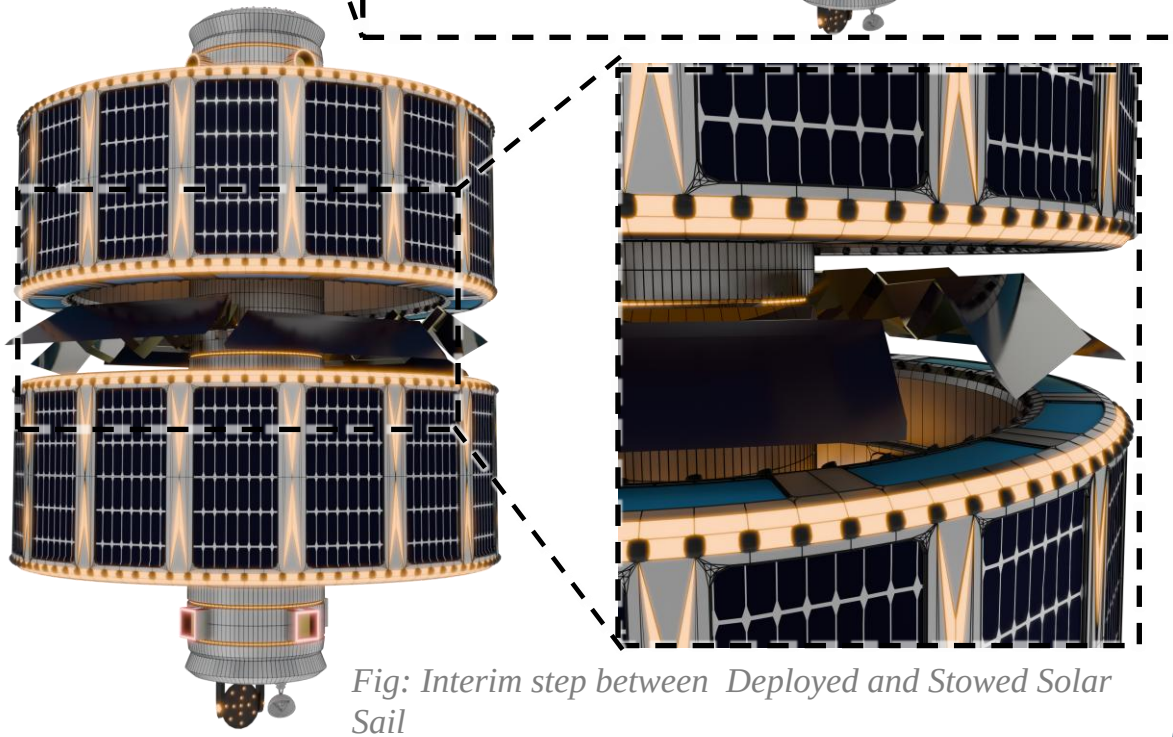


Fig: Interim step between Deployed and Stowed Solar Sail
Source: Nakshatra M.S., Blender

SPACE TUG EXTENSION

- The settlement will use space tugs to boost for maneuvering.
- The space tugs will be connected to docks present on an extended circular flat interface of 5 meters in diameter.
- The tugs, even multiple, will provide thrust at 1/10th of a g.

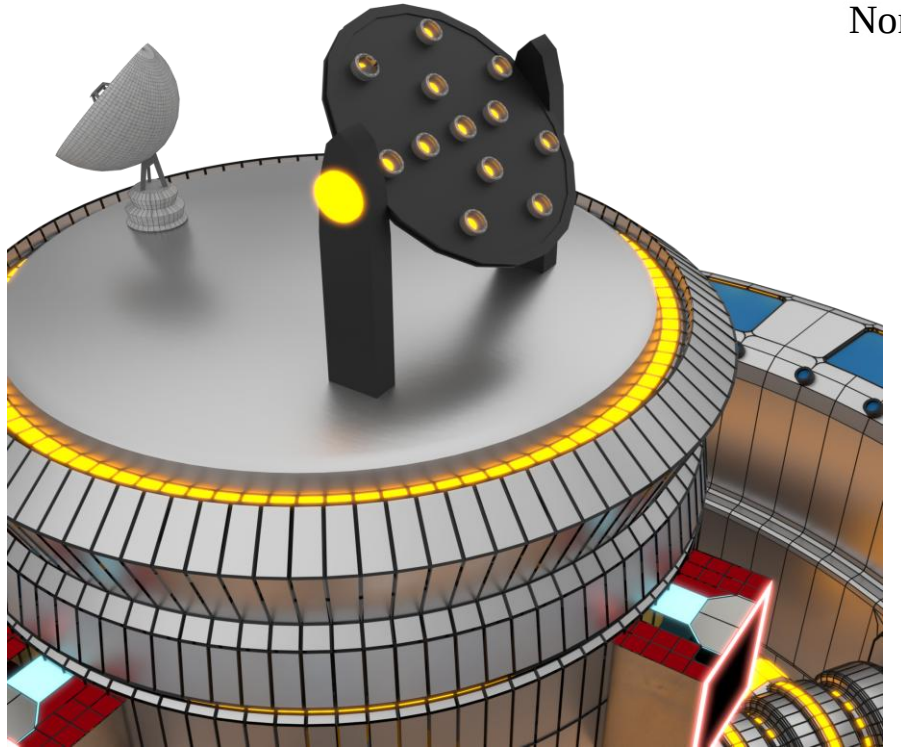


Fig: Space Tug Extension Articulation
Source: Harris Wang., Blender

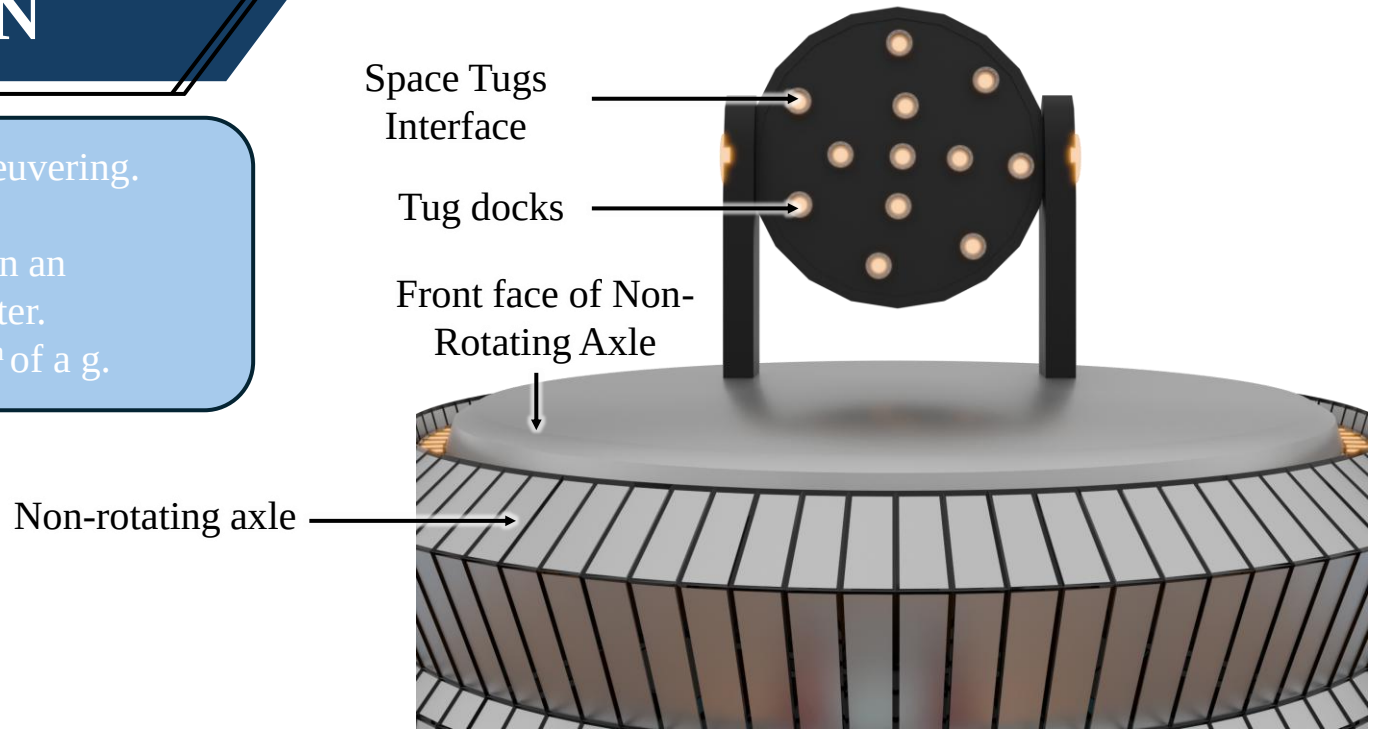
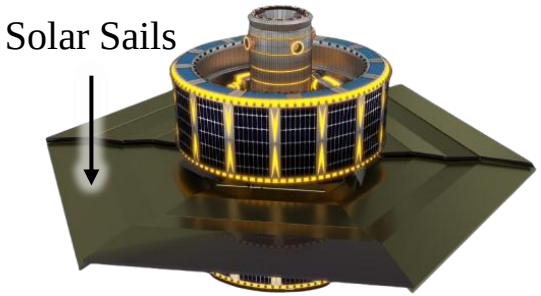


Fig: Space Tug Extension
Source: Harris Wang., Blender

- This extended interface will be present on the front of the axle and will be articulated.
- The interface will allow 270 degrees of mobility.
- This rotation will allow the space tugs to boost the settlement in all directions
- The interface will host multiple docks of the same size to which the tugs and their tethers will be attached.
- At a time, multiple tugs can be attached for increased maneuverability.

AUTOMATION IN SOLAR SAILS

STEP 1:

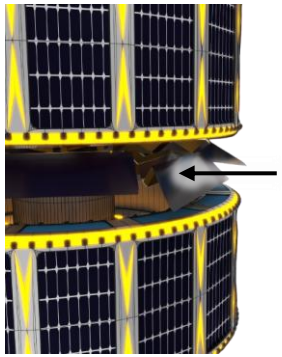


Initial Deployed Configuration:

Solar sail status: Fully deployed
Orbital position: Between Mars and Earth

Then ‘Solar sails’ are made up of **Flexible Polymer with Carbon fibre** covered with **reflective silicon**. The shape-memory polymer composites (**SMPCs**) have fully extended the booms, and the sail is actively capturing solar radiation. The **control room** will continuously monitor the position of the sail.

STEP 2:

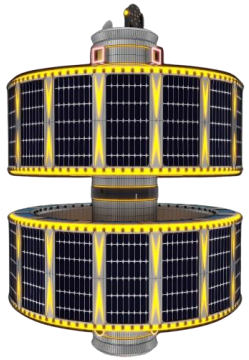


Mid Contraction:

Solar sail status: Contracted slightly
Orbital position: Close to the Mars orbit

The **trimming process** is initiated, reducing the surface area of the sail. The sail is trimmed, adjusting its **angular orientation** to generate enough thrust for velocity control. The space tug interface begins **boosting the settlement**, using the **circular flat interface**. The composite booms have begun to **fold progressively.**

STEP 3:



Stowed Configuration:

Solar sail status: Fully contracted
Orbital position: Transition phase, full cycle of the orbit

The shape-memory polymers **fully coil** the sail back into a **compact, stowed** configuration. All **locking mechanisms** are engaged, ensuring that the system can withstand the stresses of the orbit. no further deployment is needed, and all systems are checked for integrity.

AUTOMATION IN SOLAR SAILS



Fig: Solar Sail Control Center, Source: Allowyn H, Stuti P, Blender

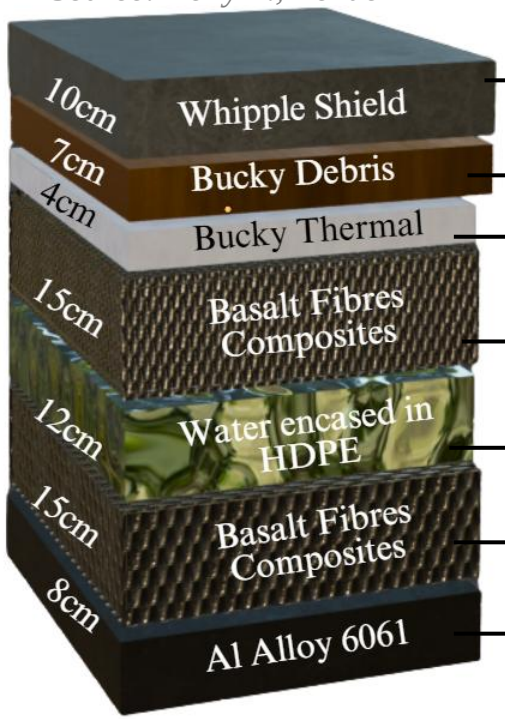
4 LIDAR sensors at the front of the axle provides a panoramic view by mapping.

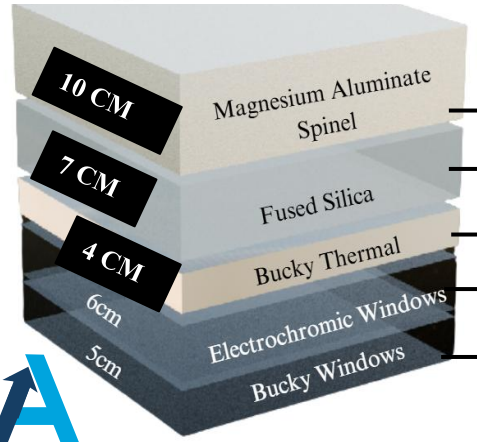
Sensors (**LIDAR, IMU**) adjust the system to avoid debris impact.

The settlement uses an array of **IMU sensors** that provide us with all the data about the status of the settlement such as velocity, location and orbit.

CONSTRUCTION MATERIALS

Fig : Construction Material, Source: Kelly Z.,Blender



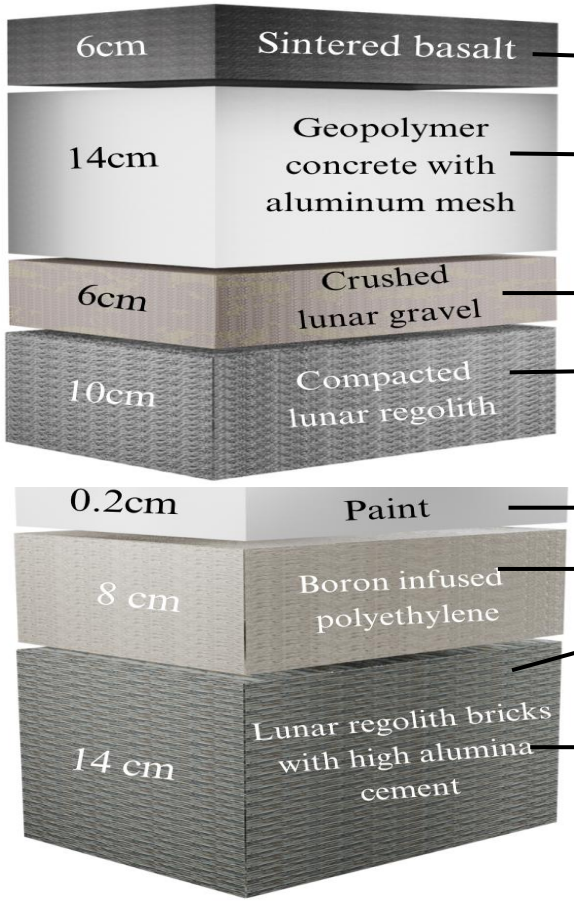


PURPOSE	SOURCING	MASS (kg)	CASSSCs
Hull Materials			
Debris Protection	Aluminium – Anorthite: Balderol	24,060,756	1526
Debris Protection	BuckyBreakthroughs	-	-
Thermal Protection	BuckyBreakthroughs	-	-
Debris Protection	Al, Si– Anorthite; Mg, Fe– Olivine (Lunar Highlands, Impact Craters): Balderol	37,400,616	2368
Radiation Protection	Water – Asteroids: Bellevistat, HDPE – ***Carbon Creators***	10,355,794	680
Debris Protection	Al, Si– Anorthite; Mg, Fe– Olivine (Lunar Highlands, Impact Craters): Balderol	37,400,616	2368
Structural Stability	Al, Si – Anorthite; Mg, Fe: Olivine (Lunar Highlands, Impact Craters); Fe, Ti: Ilmeinite; Mn, Pyrolusite (South Pole-Aitken Basin): Balderol	19,261,405	1214
Window Materials			
Debris Protection	Mg, Al, O – Spinel: Lunar highland	23,232,786	1,464
Thermal Protection	Lunar Highlands: Balderol	9,993,984	630
Radiation Protection	***Carbon Creations***	3,115,008	197
Imitates day and night cycle	Earth	9,734,400	614
Pressure Retention	BuckyBreakthroughs	-	-

Table : Construction Material, Source: Harshit A., Charlie S., Aarav K., Gaurav A., Mahum K., Microsoft PPT

INTERNAL MATERIALS

Fig : Construction Material, Source: Kelly Z., Blender



PURPOSE	SOURCING	MASS (kg)	CASSSCs
Road Materials			
Wear Resistant Layer	Lunar Basal : Balderol	4,582,490	289
Stability and magnetic interaction	Ca, Al, Si, O - Anorthite (Lunar Highland) : Balderol	23,493,090	1,469
Drainage And Distributes Load	Crushed lunar rock debris South Pole-Aitken Basin : Alaskol	7,394,189	465
Stable Foundation	Lunar Regolith - Alaskol	14,694,034	926
House Materials			
Protective Finish	***Seals-It-All***	561,593	36
Protection Againt Solar Flares	***Carbon Creations***	11,243,244	709
Structuaral Stability (Bricks)	(Lunar Highlands) : Alaskol	19,322,904	1,220
Structuaral Stability (High Alumina Cement)	Al, Si, O - Anorthite and Mg, Fe-Olivine (Lunar Highlands) : Balderol	3,780,400	239

Table : Construction Material, Source: Harshit A., Lucian D., Mahum K., Taha R., Microsoft PPT

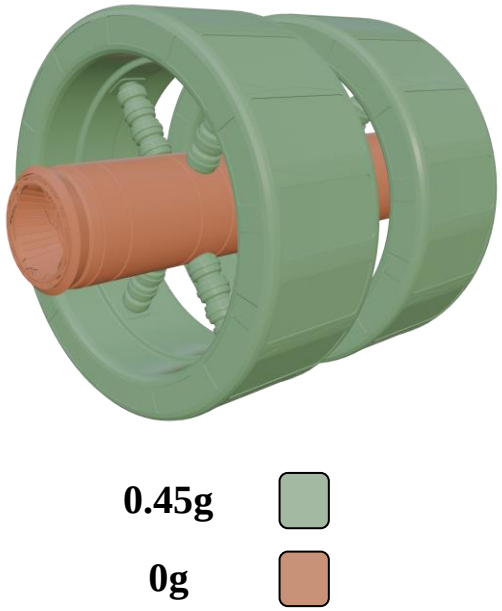
Solar Sails

Lunar Ores from Balderol will be sent for to Bellevistatmined and refined Lunar Regolith from Alaskol will be transported to the Benevectoras construction site through the mass driver present at the Lunar Equator, which is subcontracted to ***Magnetic Propulsion Company***.

ARTIFICIAL GRAVITY

Through the entirety of the structure both the rings will be pressurized and axle unpressurized.

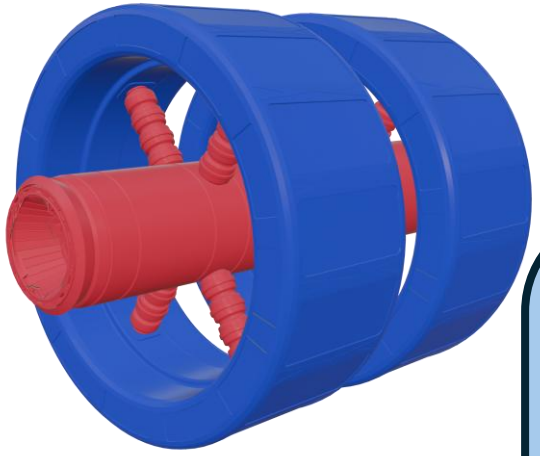
The settlement will have 2 artificial gravities levels: **0.45g** in the rotating rings and **0g** in the axle.



0.45g

0g

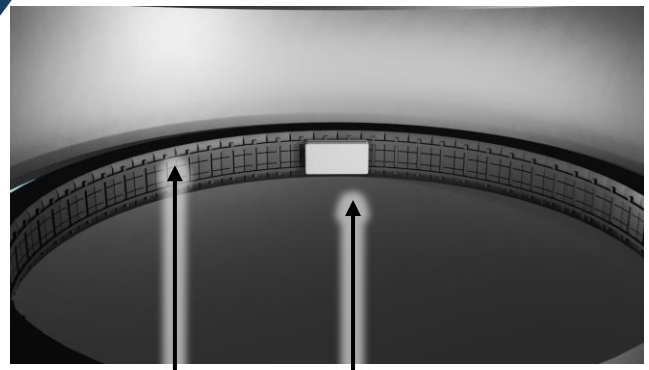
Fig: Rotational and Non-rotational
Source: Sumay M., Blender



Unpressurized

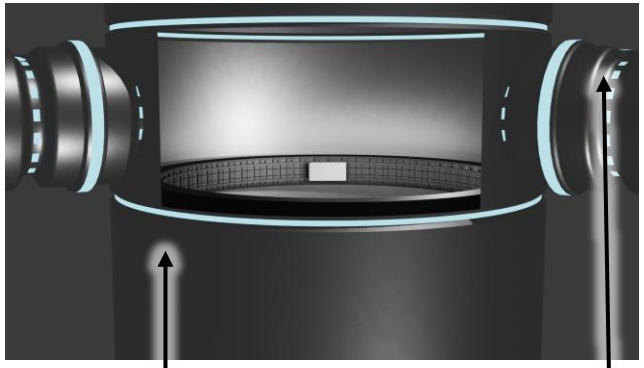
Pressurized

Fig: Pressurized and Unpressurized
Source: Sumay M., Blender



Pod Rails

Pod



Axle

Spokes

Connection Apparatus and Rotational Interface

- The connection apparatus will consist of spokes, connecting the non-rotating axle to the rings. Ball bearings will be present at the connection to ease friction.
- The spokes will also contain pod rails for the movement of the pods which are a part of the rotational interface.
- The rotational interface will consist of a pod mounted on tracks on the inner circumference of the non-rotating axle.
- These tracks will be right beside the opening to the spokes which will be rotating.
- The pod can accelerate or deaccelerate on the tracks to match the speed of the rotating spokes to transfer cargo or passengers from the pod to the pod in the spokes, thus creating a transfer method.

DOCKING & WAREHOUSING

DOCKING

- There will be 2 ports on opposite sides of the axle. 1 for cargo (square), named **Helios** & 1 for people (circular), named **Aether** each with 4 docks.
- All docks will use **androgynous docking systems**.
- They will be pressurized with airlocks subcontracted to ***Lossless Airlocks.***

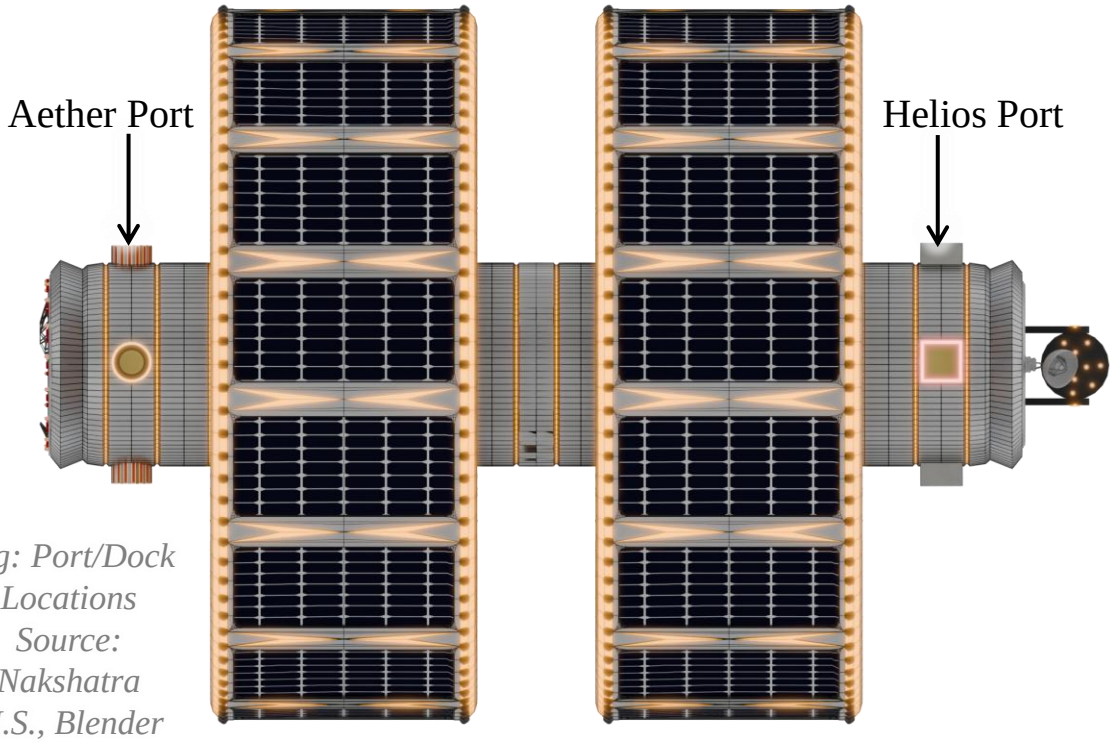


Fig: Port/Dock Locations Source: Nakshatra M.S., Blender

Androgynous docks will accommodate a **variety** of spacecraft and facilitate **quick** docking and undocking.

WAREHOUSING

An unpressurized warehousing volume will be present at **Benevectoras** in the axle with a **capacity of 864 CASSSCs.** It will house the following things:

- **Harvested** Martian Materials
- Commodities
- Items for **Export & Import**
- Miscellaneous Items

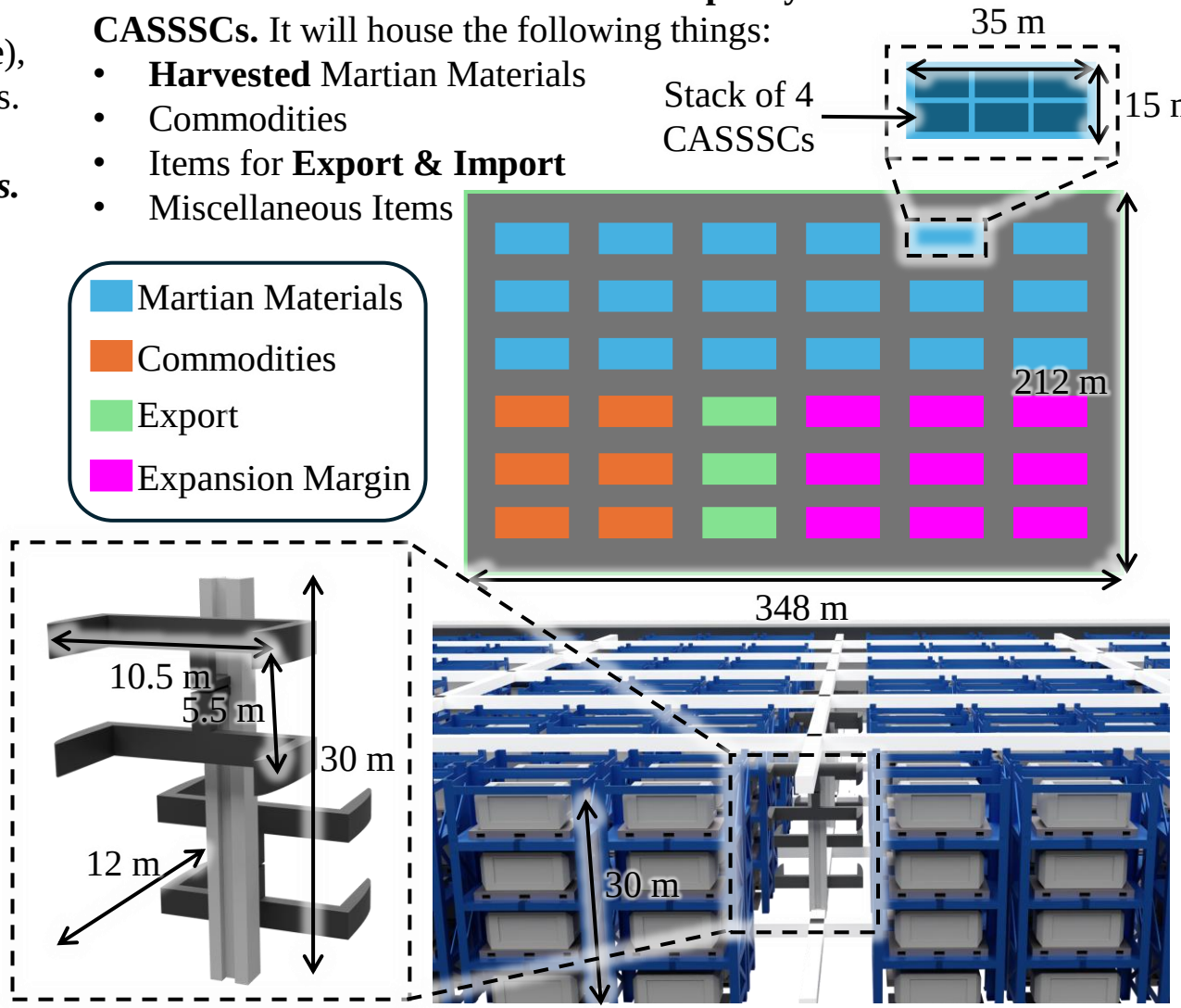


Fig: Warehouse Floorplan & Render & Forklift Render Source: Ali Abbas R., Blender

Autonomous floor & roof mounted forklifts will manage the warehouse.

CARGO MANAGEMENT

LOADING CARGO:
After a spacecraft docks, the **robotic arm** present inside will load the **CASSSC directly into the CTM**. 1 CTM will have a capacity of 8 CASSSCs.

BEING STORED:
The CTM will **ship the CASSSC(s)** to the warehousing volume present in the central axle. It will be able to **take CASSSCs from multiple docks** prior to going to the warehouse.

FORKLIFTS:
Autonomous floor & roof mounted forklifts will unload the CASSSCs from the CTM at the Loading & Unloading Bays (LUBs).

BEING EXPORTED:
CASSSCs will be used to store all items from the industrial park. CTMs will take those items from the industrial park to the warehouse or straight to the docks. The same applies to items from the Research Labs.

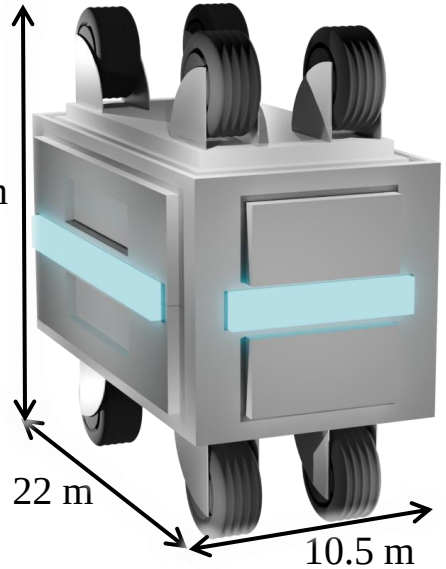


Fig: Cargo Transport Module (CTM) & Channel
Source: Ali Abbas R., Blender



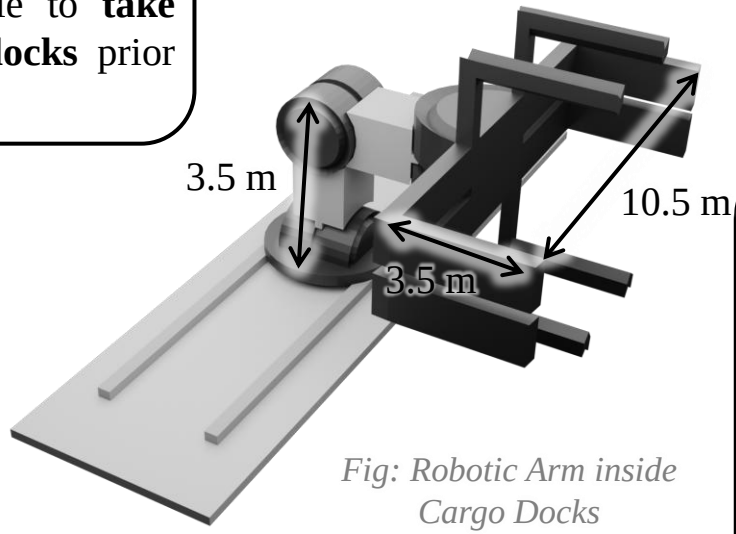


Fig: Robotic Arm inside Cargo Docks
Source: Ali Abbas R., Blender

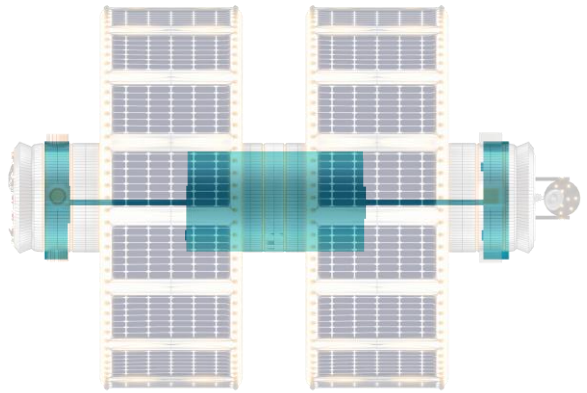


Fig: CTM Channel Routing
Source: Nakshatra M.S., Ali Abbas R., Blender

Table: CTM Characteristics
Source: Ali Abbas R., Microsoft PowerPoint

Characteristic	Qty
Number of CTMs	10
Capacity	8

ENTRY/ EXIT SYSTEM

Entry/Exit systems will include 2 types of airlocks: **Vehicle and Cargo airlocks**

- Vehicle airlocks will be used by passenger and passenger vehicles**.** It will have a specific height so as to accommodate the size of the vehicles.
- Cargo airlocks will accommodate vehicles, up to the size of one CASSSC.

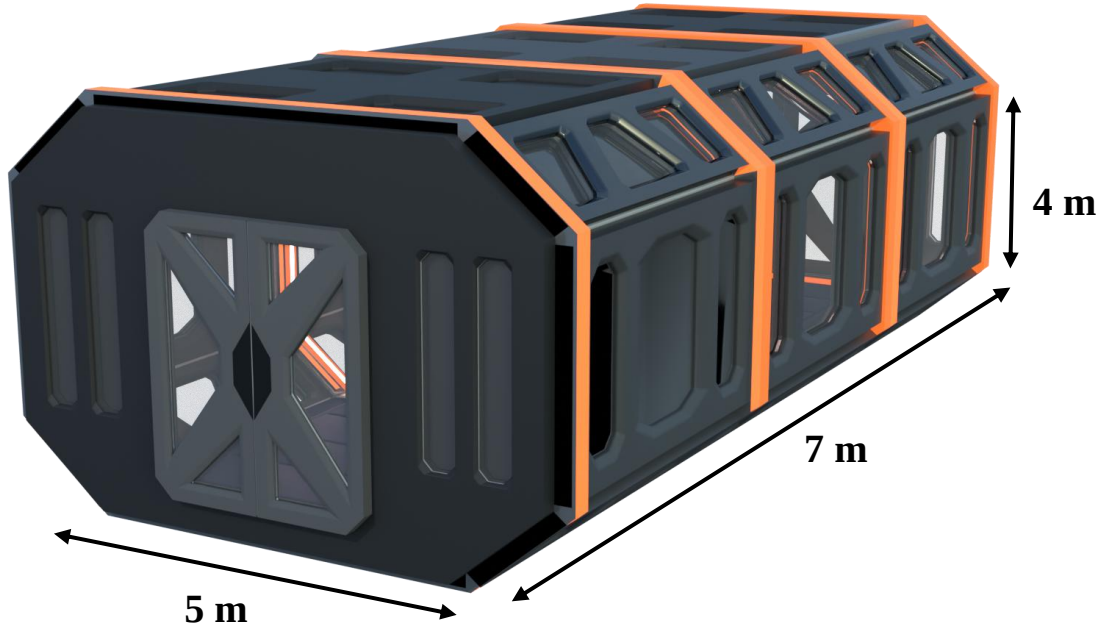


Fig: Passenger Airlocks
Source: Krishav A., Blender

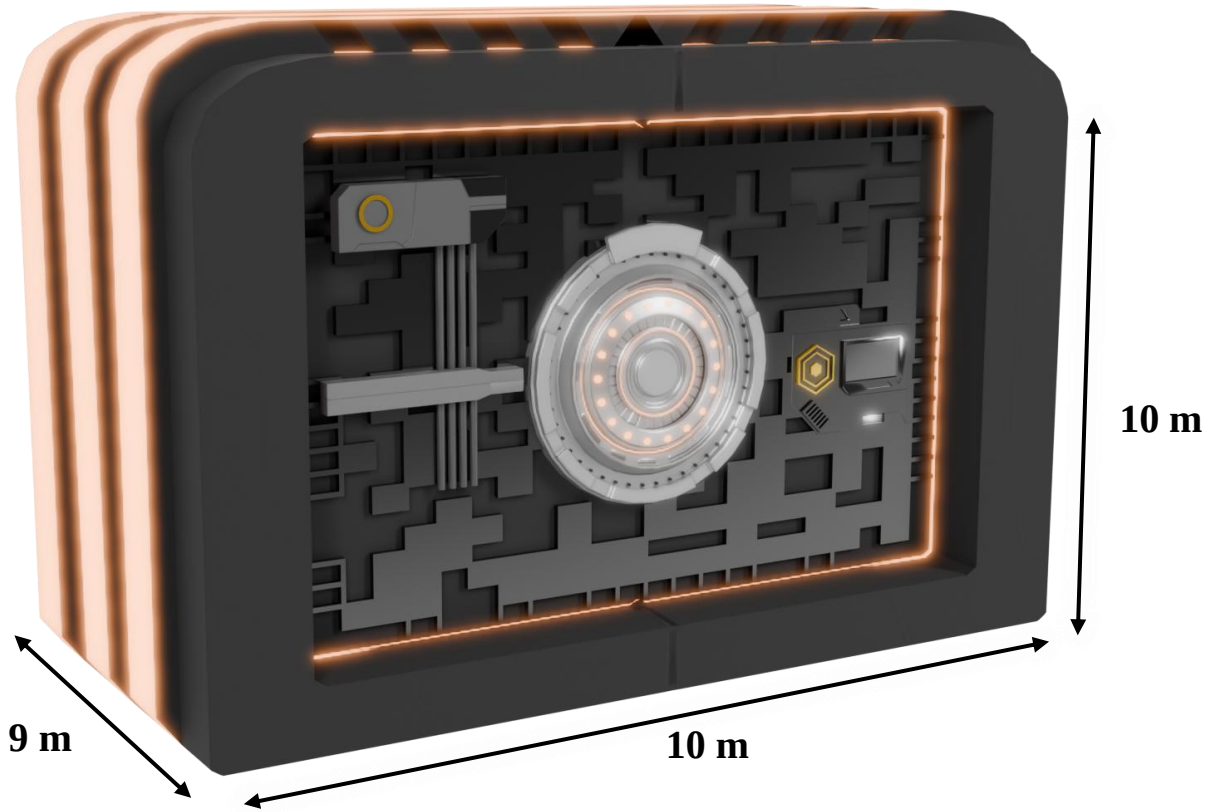


Fig: Cargo Airlocks
Source: Harris W,., Blender

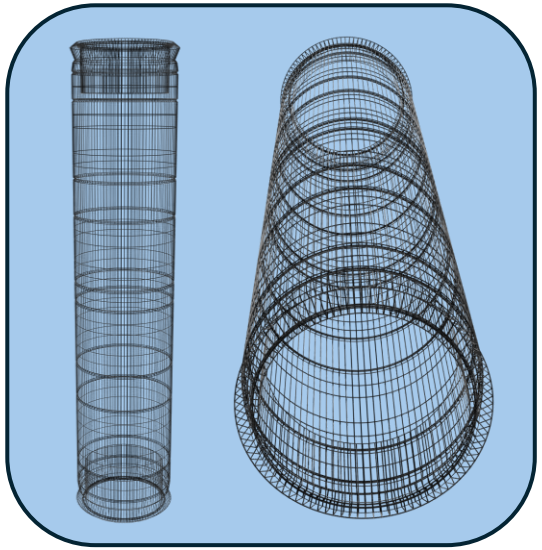
CONSTRUCTION PROCESS

1



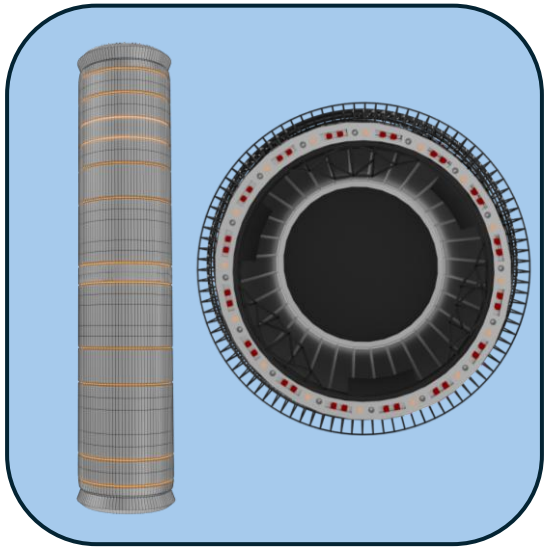
- A construction shack will be utilized for initialization of construction.
- Construction will begin at Earth-Moon L1 Point.

2



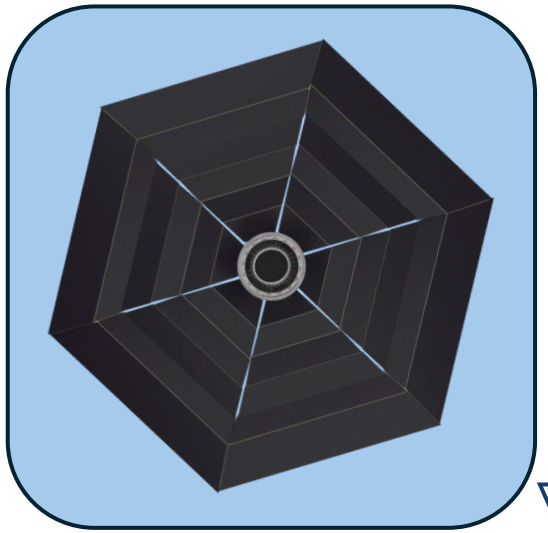
- Trusses for the non-rotating axle will be laid down, followed by the materials, effectively finishing the exterior. CASSSCs will be repurposed here.

3



- The expansion latch system will be constructed and sealed off till it needs to be used.

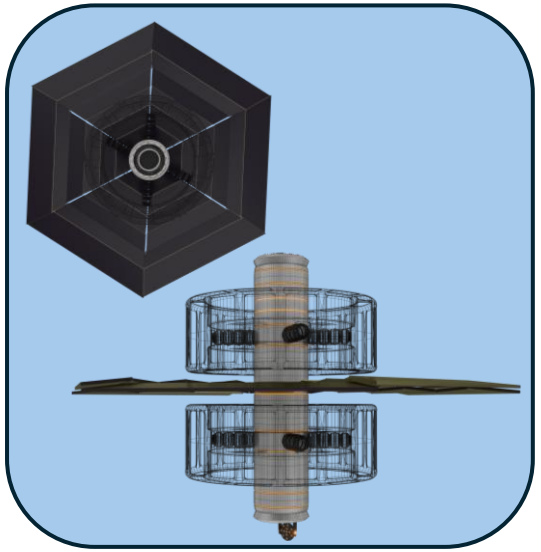
4



- A system for the deployable solar sails will be laid down, which allows advanced mobility and storage of the sails in their stowed-form.

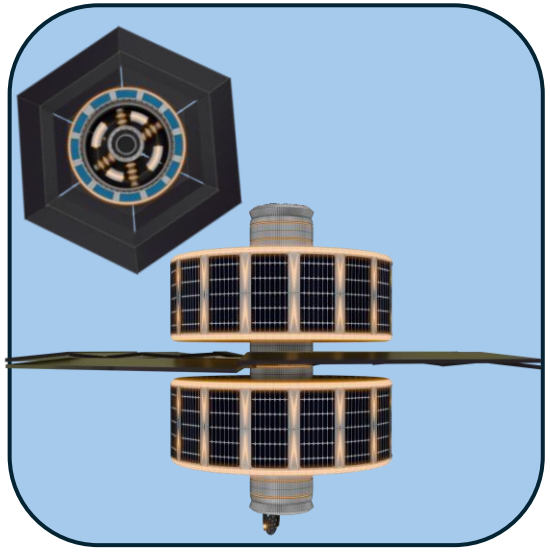
CONSTRUCTION PROCESS

5



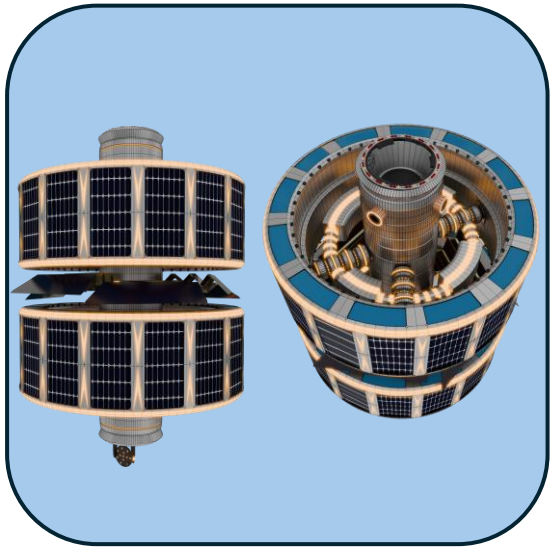
- Trusses and materials for the space tug interface will be laid down, with trusses for the rotating rings

6



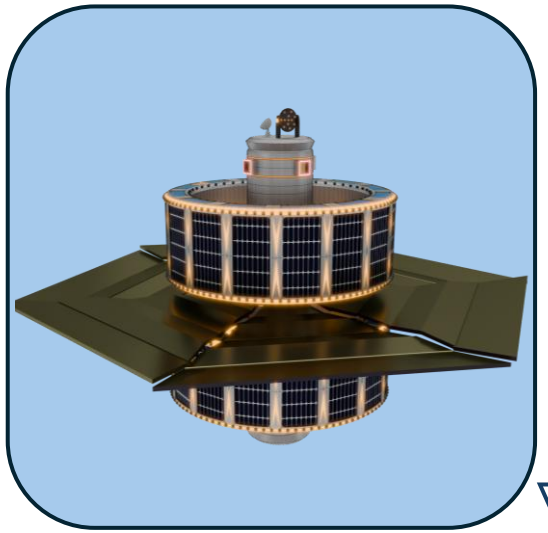
- Materials will be laid on all the remaining trusses, thus completing the space tug interface and the rotating ring exterior.

7



- Two tori will be built on the inner circumference of the rings, and will serve as research labs. These will be connected through passages.

8



- Communication modules subcontracted to ***OrbitLink Communications*** will be installed, rendering the structure complete.

INDUSTRIAL & RESEARCH

INDUSTRIAL VOLUMES

Industrial volumes will be located on the **spokes** of the settlement. **Each industrial section on each spoke will consist of 3 sectors,** each with 1 manufacturing and 1 processing facility.

- Gravity level of **0.3 g.**
- CASSSC repurposing.
- Produces products for export and **products made of Martian materials.**
- Successful **products from the research labs** will also be produced here.

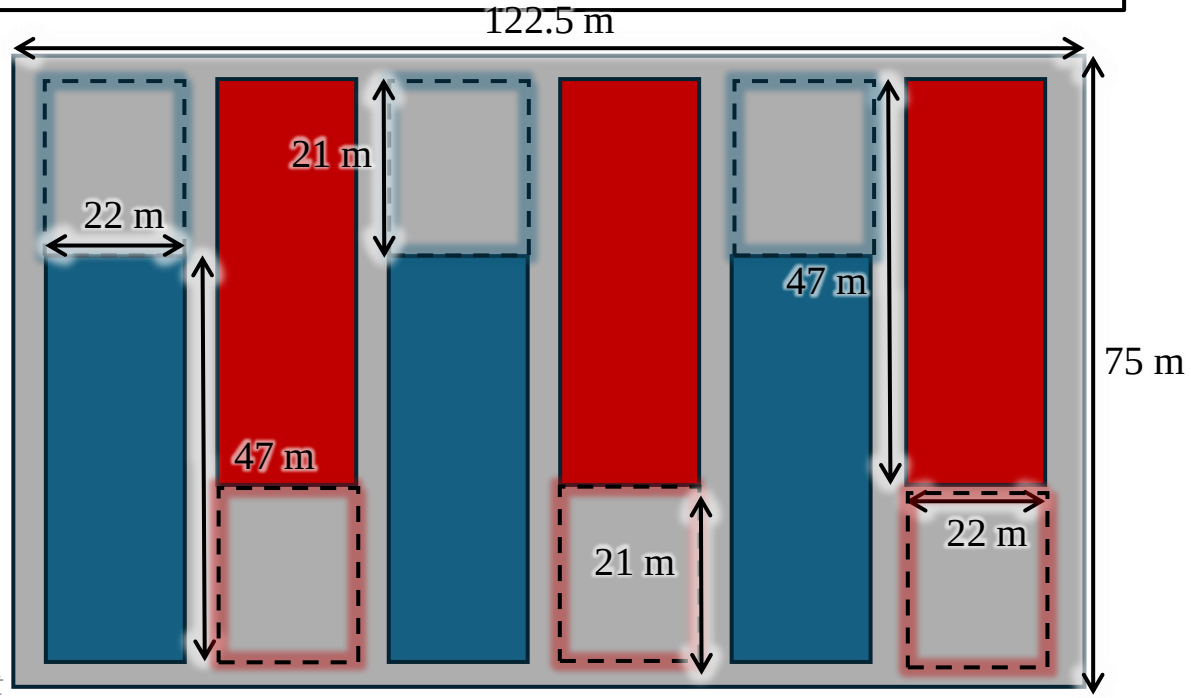


Fig: Industrial Park Floorplan Source: Ali Abbas R., Microsoft PowerPoint

RESEARCH LABS

Observation Decks: Where researchers **observe** the testing chamber.

Workstations: A place for researchers to **perform small tests/experiments.**

Research stations: A place for researchers to **analyze and collect data.**

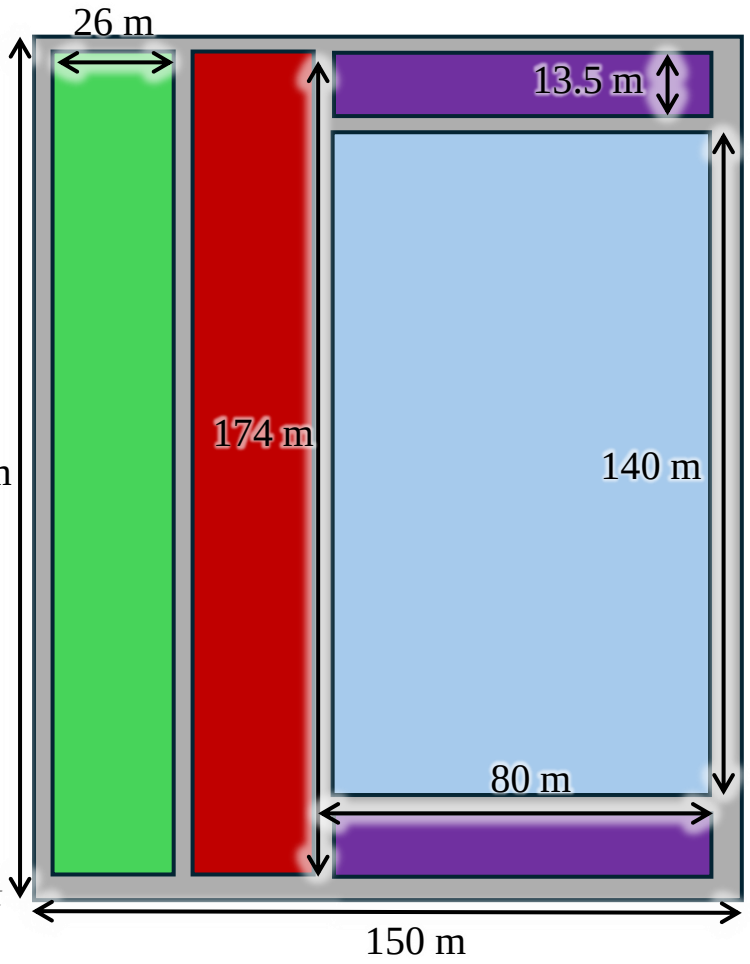
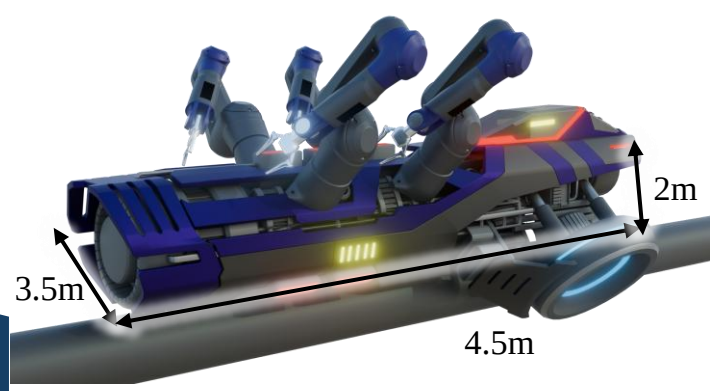


Fig: Research Lab Floorplan Source: Ali Abbas R., Microsoft PowerPoint

Testing chamber: Here **large-scale experiments and tests on materials** will be carried out in variable pressures simulated with airlocks subcontracted to ***Lossless Airlocks.***

AUTOMATION IN CONSTRUCTION



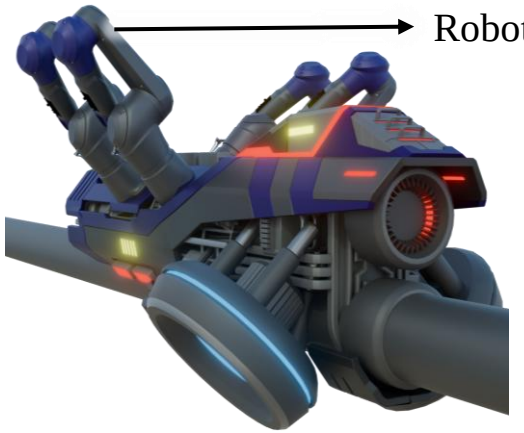
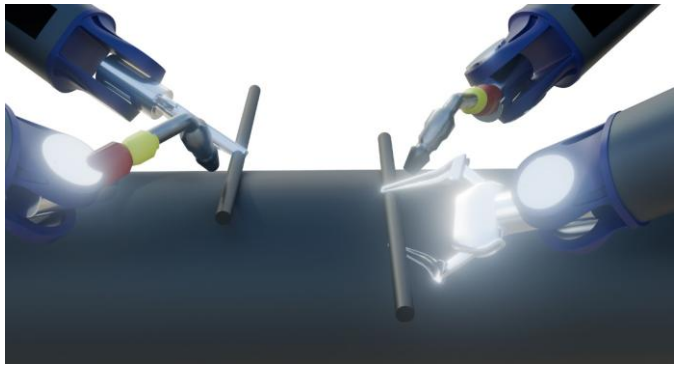


Fig: ECB, Source: Yatharth G., Blender

SENSORS	LIDAR, IMU, Proximity, Thermal
POWER SOURCE	Primary: Solid Oxide Fuel Cells (SOFC) using methane Backup: Silicon Graphene batteries
MATERIALS	Aluminum 6061, Silicon Buckystructure (Thermal Variant), UHMWPE, Debris-protective Silicon Buckystructure, *Electroprotect* for internal circuitry, Faraday Cage from carbon nanotubes subcontracted to *TubularTechnologies*
MACHINERY	MIG welder, TIG welder, Adhesive , Cordless Drill, Sander, Circular Saw

Table: ECB Specs, Source: Stuti P, Angela W., PPT



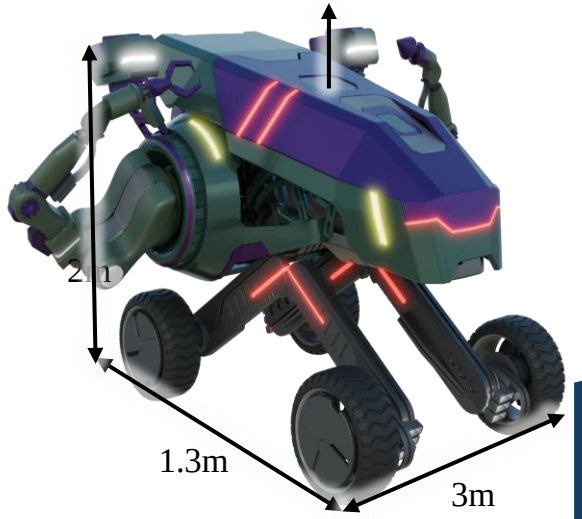


Fig: ICB, Source: Yatharth G., Blender

SENSORS	LIDAR, IMU, Proximity, Tactile, Pressure, Infrared
POWER SOURCE	Primary: Silicon Graphene batteries
MATERIALS	Aluminum 6061, Silicon Buckystructure (Thermal Variant)
MACHINERY	MIG welder, TIG welder, Adhesive , Cordless Drill, Sander, Circular Saw

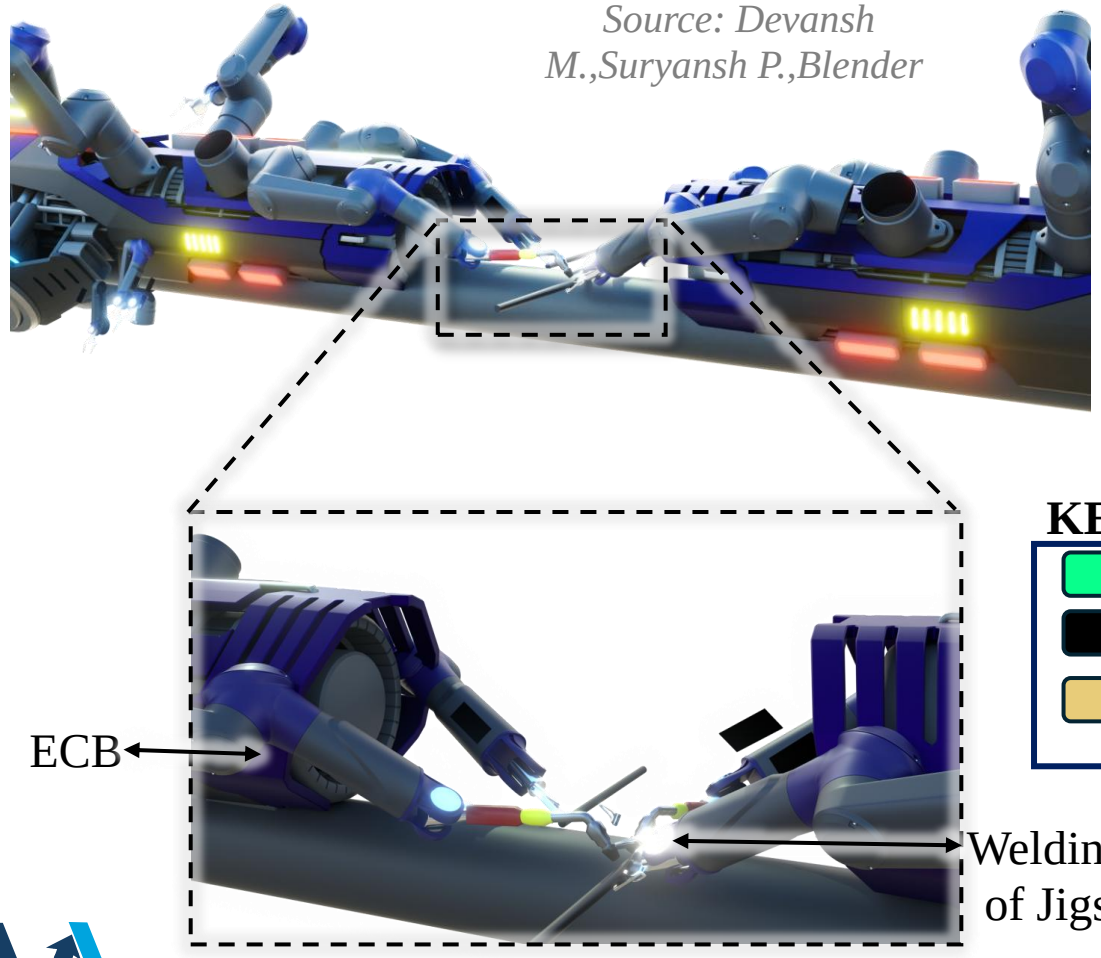
Table: ICB Specs, Source: Stuti P, Angela W., PPT

Self-programmable "**rollerbots**" for the **wheels** of the bots will be provided by **ROCKDONNELL** via *Litigation Limiters*. Multi-purpose "**F1M8**" robots for the ICB and the ECB will be provided by **GRUMBO AEROSPACE** via *Litigation Limiters*.

JIGS AND TRASPORTATION

- Jigs will form the **exoskeleton** of the structure
- They will be **uniform** throughout the structure

Fig: Assembly of Jigs
Source: Devansh M.,Suryansh P.,Blender



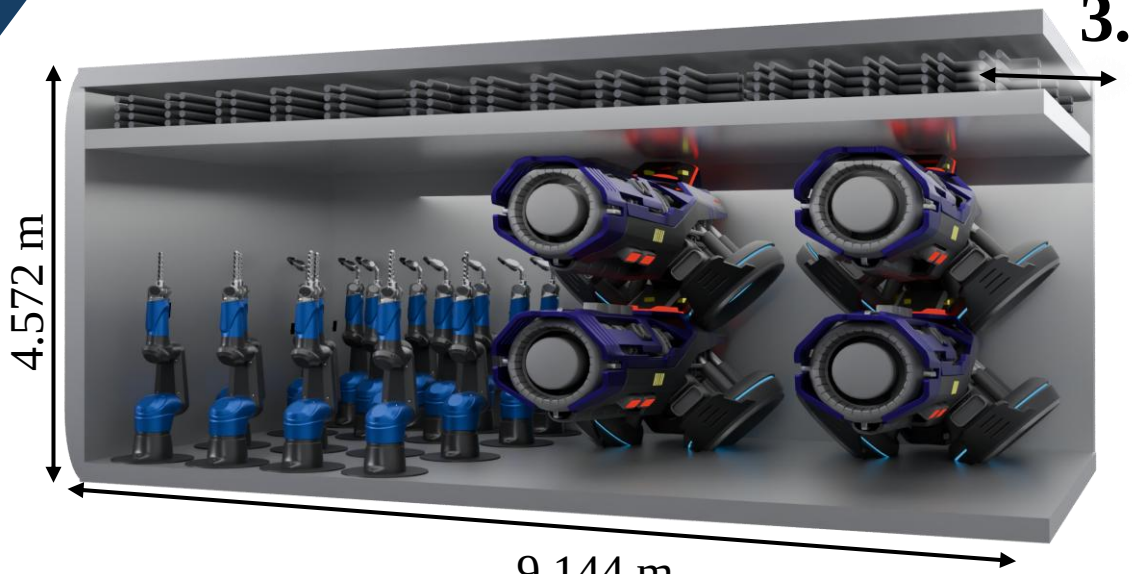


Fig: Transportation of Jigs,Machinery&Bots
Source: Devansh M., Blender

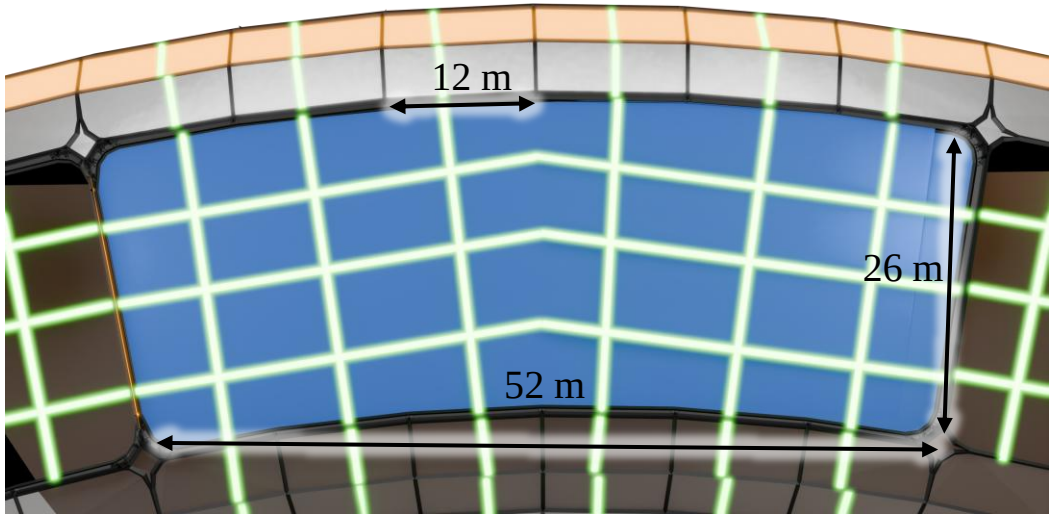
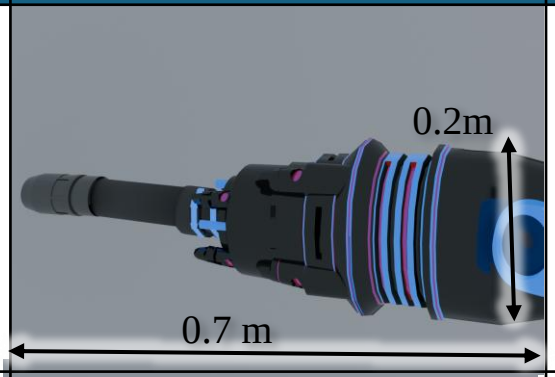
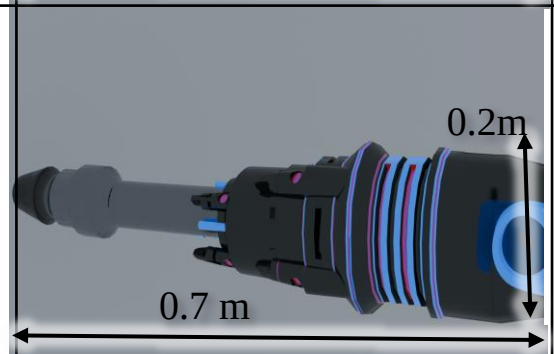
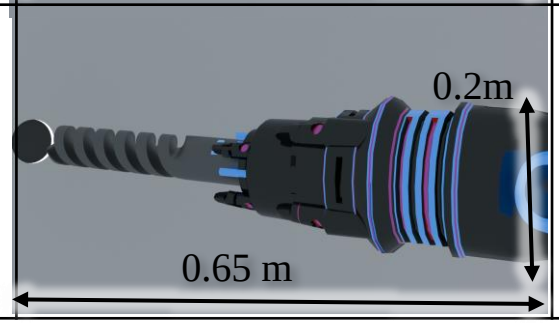


Fig: Jigs
Source: Devansh M., Blender

PRIMARY MACHINERY

Machinery	Function	Illustration	IOC/OC
MIG WELDER	Joins metal with wire electrode and gas.		Initial Operatio-nal Cability (IOC)
TIG WELDER	Precision welding with tungsten electrode. .		Operatio-nal Construct-ion (OC)
CORDLESS DRILL	Drills and fastens.		IOC and OC

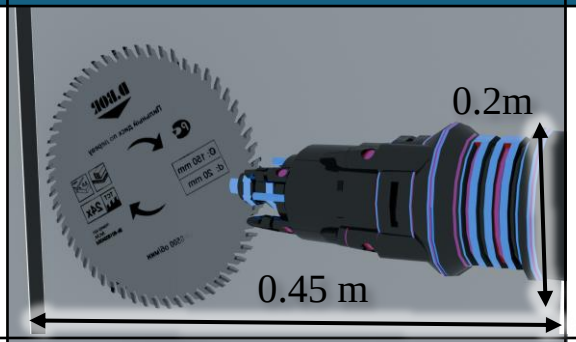
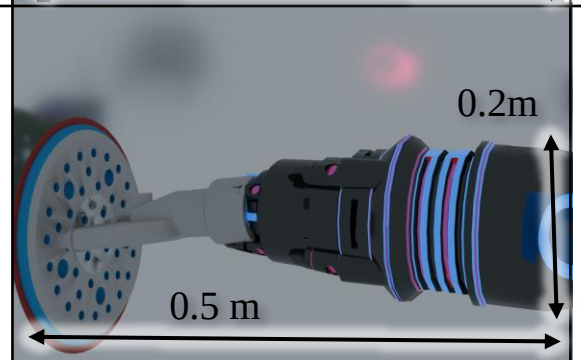
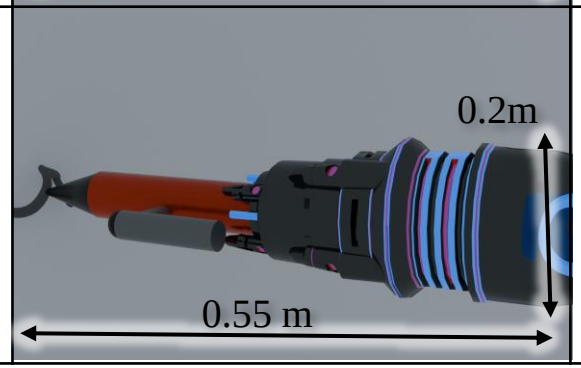
Machinery	Function	Illustration	OC/IOC
CIRCULAR SAW	Cuts metal or plastic.		IOC and OC
ORBITAL SANDER	Smooths surfaces.		Operatio-nal Construct-ion (OC)
ADHESIVE GUN	Bonds materials chemically		Operatio-nal Construction (OC)

Fig: Primary Machinery

Source: Devansh M.,Mahum K.,Microsoft PPT

INTERIOR LAYOUT

- Each ring will have five sectors allocated for residential, commercial, and agricultural purposes, with a consistent vertical clearance of 40 meters.
- Windows will be provided for clear views of space and Earth and Mars when they're nearby.
- The windows will feature transparent displays, offering informational overlays or transforming into an immersive entertainment system.

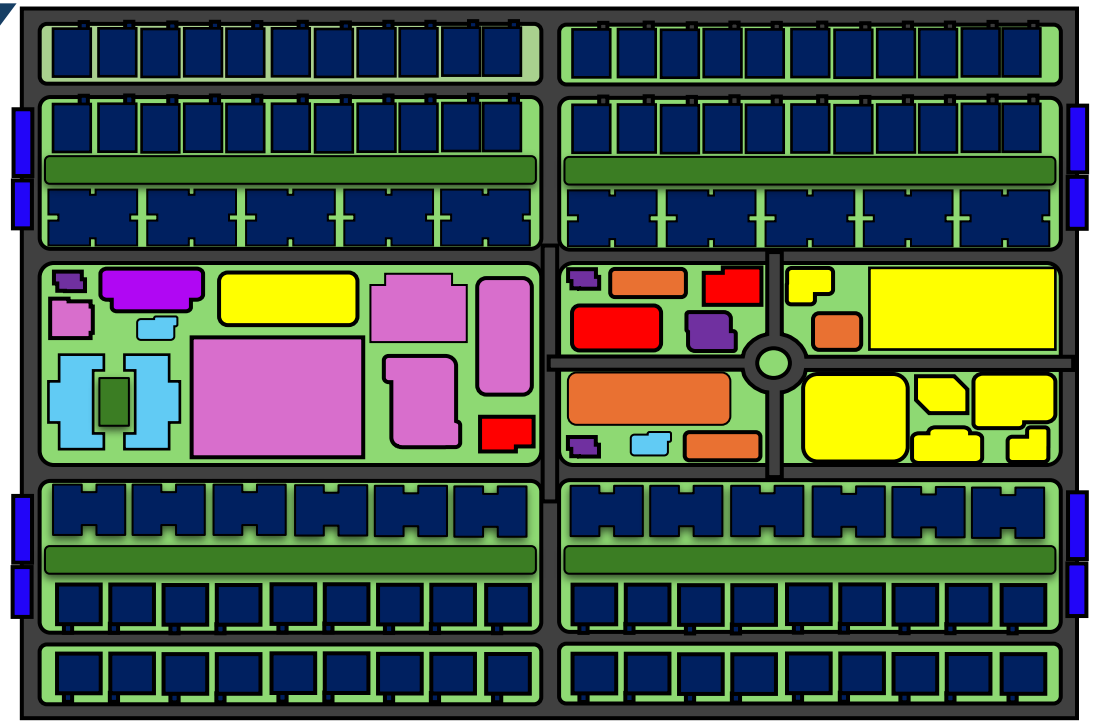


Fig: Interior Layout of a Sector
Source: Aatreyi S., Vihaan R., Microsoft PPT

TOTAL AREAS IN BENEVECTORAS

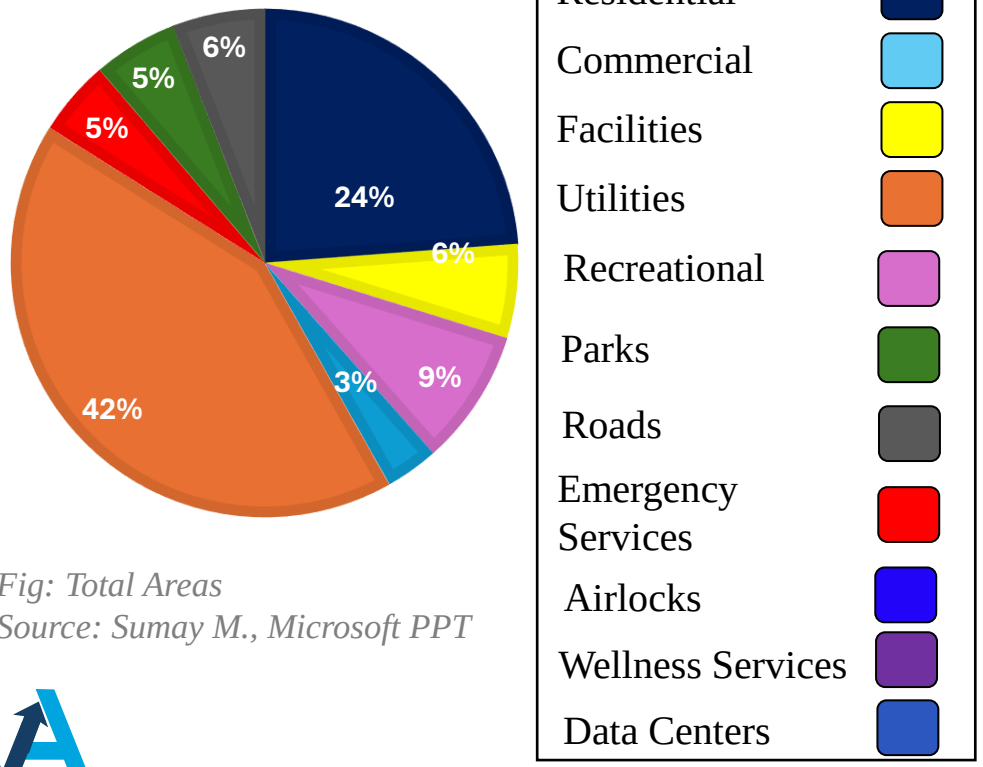


Fig: Total Areas
Source: Sumay M., Microsoft PPT

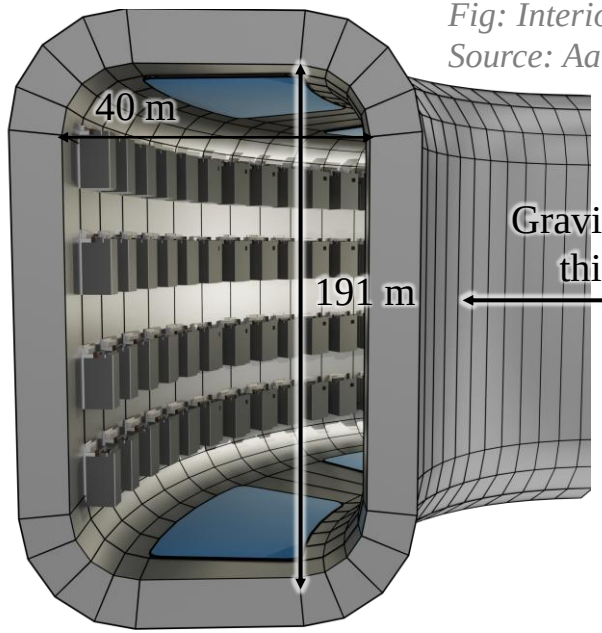
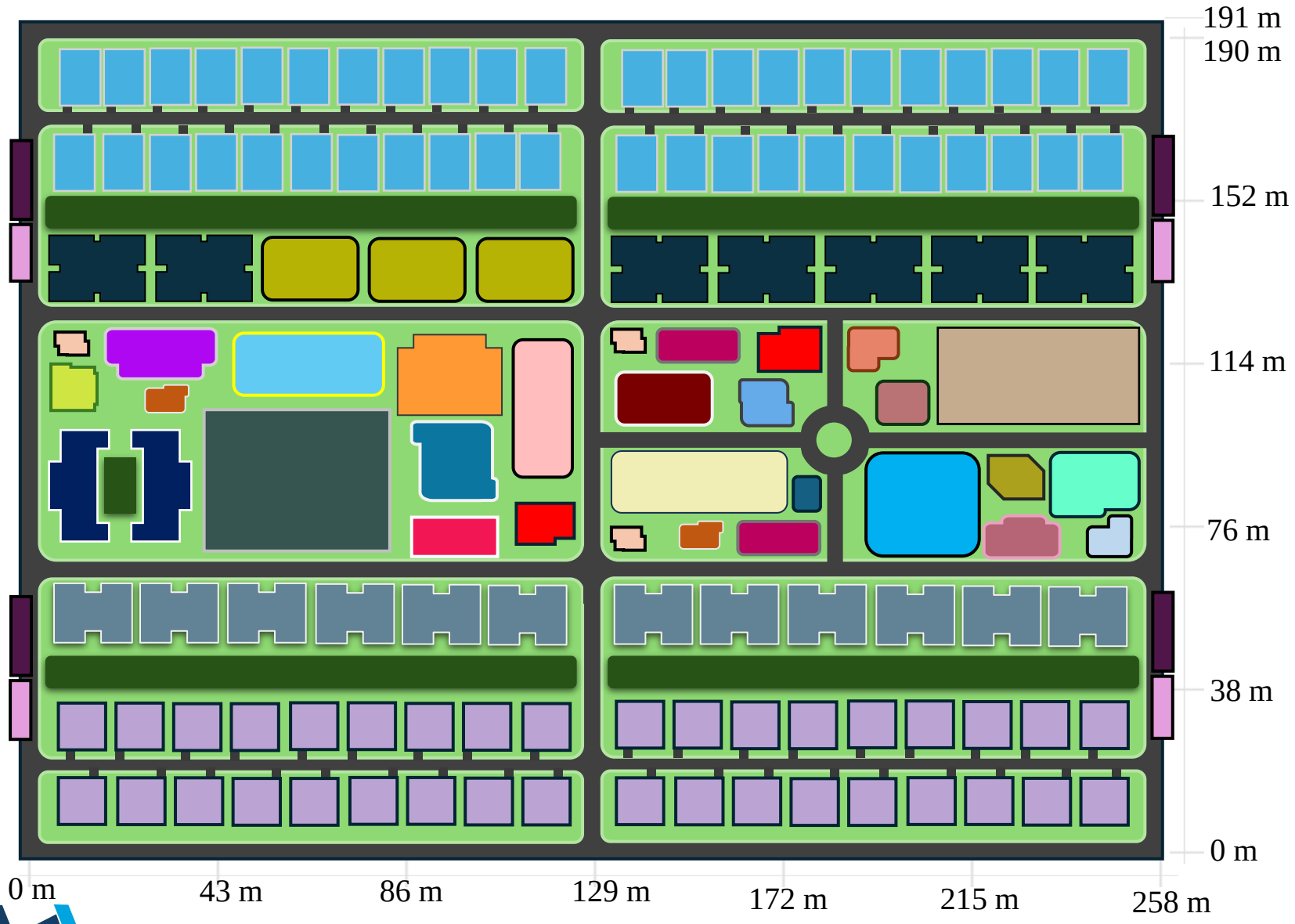


Fig: Orientation of Down Surface Area
Source: Nakshatra M.S.., Microsoft PPT

COMMUNITY DESIGN

Fig: Community Design
Source: Aatreyi S., Microsoft PowerPoint



- 2 Bedroom Apartment
- 1 Bedroom Apartment
- Individual House Type 1
- Individual House Type 2
- Medical Clinics
- Haven & Health
- ImmunoZone Hubs
- Conversion Building
- Landfill Building
- Public Safety Division
- Fire Station
- Sports Facility
- Entertainment Arena
- The Creative Circuit
- Bottom Cleaners
- Air Tank
- Water Tank
- Hotel
- Bot Storage
- Control Centre 1
- Control Centre 2
- Aeroponic Facility
- Offices
- Community Centre
- Microgravity Wellness Centre
- Retail District
- Horizon Stations
- Cargo Airlock
- Personnel Vehicle Airlock
- Restaurants
- Parks
- Educational Centers
- Data centres

All roads will be designed with **gentler curves** to minimize sharp turns and mitigate the effects of the Coriolis force.

FACILITIES

ENTERTAINMENT

An **Entertainment Arena** with trampoline parks, **laser tag arenas**, and **roller coaster-like experience** where magnetic tracks and low gravity create loops, spins, and weightless moments.

The **Community Center, with multi-purpose halls and event spaces** will serve as a vibrant hub for **social gatherings** and **cultural events.**

The **Astral Dome** will serve as a **space exploration lounge**, featuring **projected simulations of the aurora borealis** enhanced with AR/VR technology, while offering **clear, unobstructed views of space.**

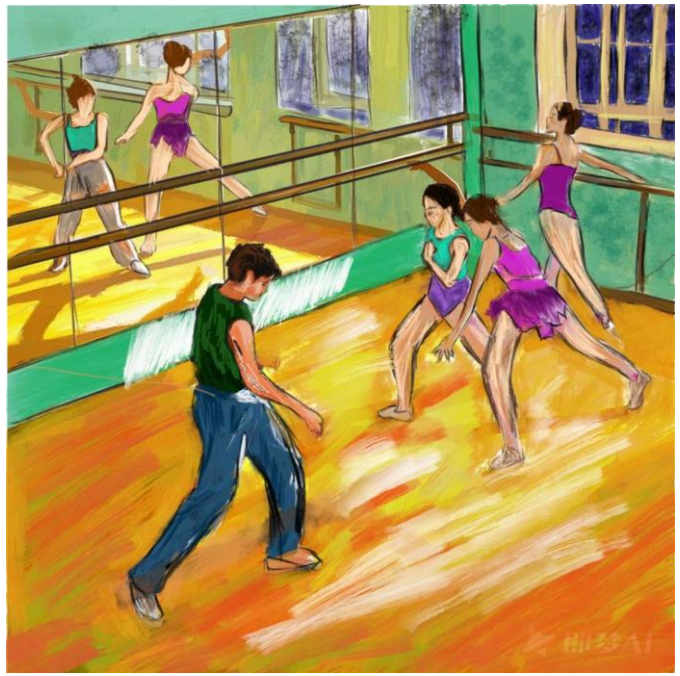


Fig: The Creative Circuit (Dance Facility)
Source: Jessie L., Notein

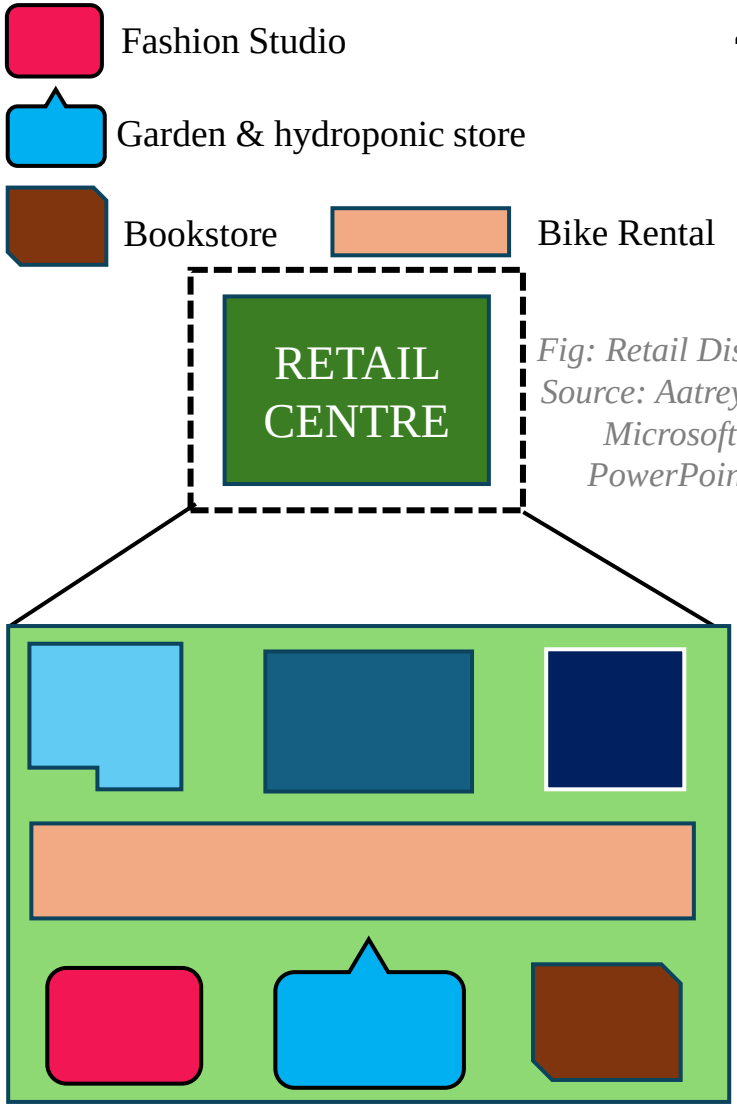


Fig: Retail District
Source: Aatreyi S., Microsoft PowerPoint

The **Educational Centers** will equip transit Mars settlers with essential skills for adapting to life on Mars.

The **office and workspace sector** will feature **rentable workspaces, incubation centers, and start-up accelerators**, directly linked to **retail areas** to support shops and commercial growth along with energy pods with guided relaxation and light therapy

FACILITIES

RECREATIONAL

Parks will include **Bloom & Brush**, botanical gardens with **outdoor art galleries** where residents can **create and display their art** while being surrounded by nature in an open-air setting.

The Creative Circuit will be a **creative hub** featuring **art studios, dance studios**, **music studios**, and **collaborative spaces**.

The **gardens and parks** will provide open spaces and feature botanical gardens to allow residents to reconnect with nature.

Haven & Health will be a tranquil environment focused on **holistic wellness**, featuring **yoga pavilions**, **meditation gardens**, and premium **spa** amenities.



Fig: ImmunoZone Hubs
Source: Karman B., Annie Han, Blender



Fig: Bloom & Brush
Source: Karman B., Blender

WELLNESS

Multiple medical clinics and **a Sports facility** with gyms, specialized training arenas, and sports areas.

The **Microgravity Wellness Center** will help counteract **bone density loss, muscle atrophy, and mobility challenges** in microgravity. With the help of **specialized exercise systems, resistance training, and targeted therapies.**

ImmunoZone Hubs, repurposed from CASSSCs, will feature **immunity-boosting bars** with **antioxidant-rich shots** and nutritious smoothies, alongside restaurants, cafés, and bars, will provide diverse dining opportunities.

MICROGRAVITY PROVISIONS

We will be using **bioengineered microorganisms** from **ROCKDONNELL**, delivered via **litigation limiters**, as an **internal source** in our body for **calcium production to prevent spaceflight osteopenia.**

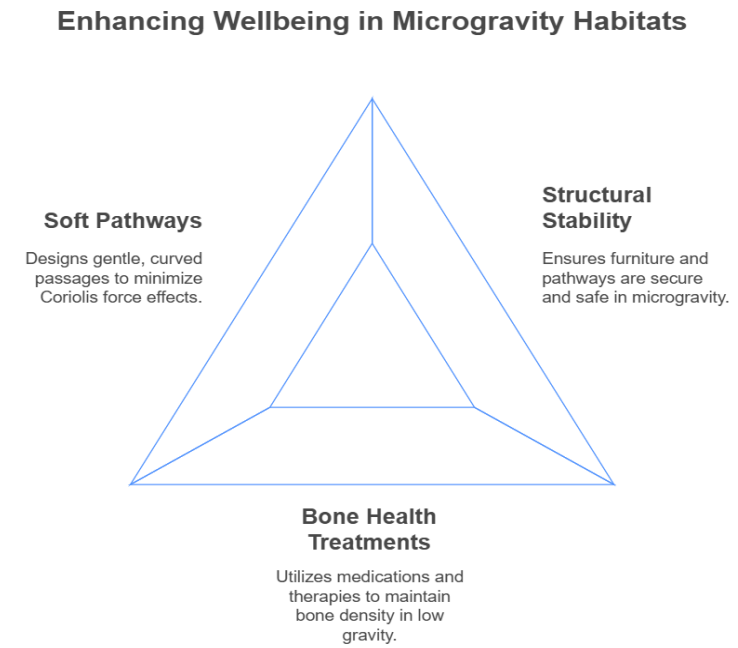
Being in such a low gravity causes bone loss and faults in our bone density. We will be providing multiple medications and treatments for the utmost wellbeing of our residents.

1. Bisphosphonates will be primarily used to **prevent and treat bone loss** by slowing down the activity of **osteoclasts**
2. **Parathyroid hormone (PTH) analogues** will be used to **stimulate new bone formation** and increase bone density.
3. Vitamin D & Calcium Supplements will be provided to residents frequently to make their health their utmost priority.

For the prevention of stages of muscle dystrophy, we will be providing well equipped fitness centers to prevent muscle loss by cardio and promote frequent exercise to prevent changes in the bone structure.

Our residents will be provided with shoes names “moonwalker” to assist them in being comfortable dealing with a lower gravity in the residential area. The shoes will contain electromagnets to help stimulate a comfortable gravity ,level for the residents.

There will not be any sharp turns in the complans to prevent effects of the Coriolis force. We will be having soft terms to accommodate for passages.



Furniture will be anchored to the ground with the help of silicon buckystructure (adhesive variant)

AIR COMPOSITION

ATMOSPHERE COMPOSITION

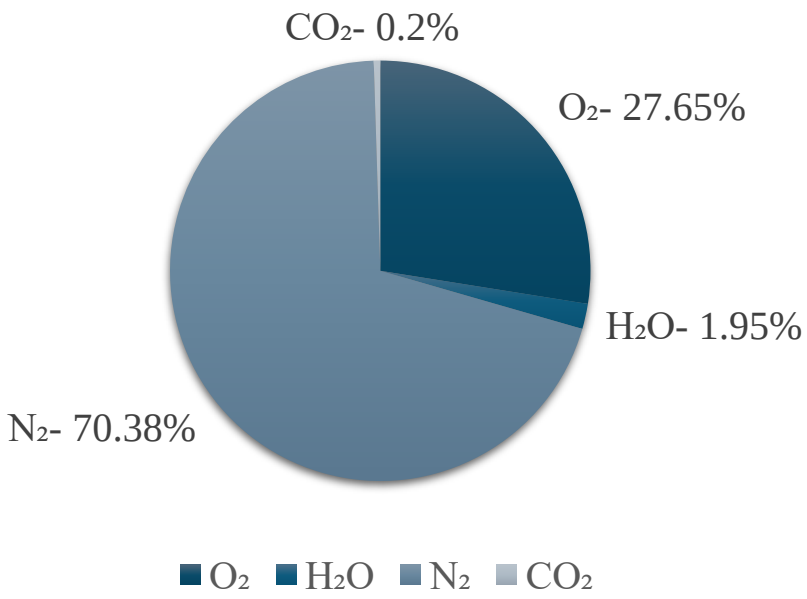


Fig: Air Composition
Source: Geet J., Microsoft Excel

Subcontracted to ***Stuff Of Life*** **(in a liquified state)**

Air revitalization will be done by ***Clean Up Your Act***

Air pressure- **0.8 Earth Atmosphere**

2 backup air tanks of **m²** per floor made by ECB from **repurposed CASSSCs**

HVAC System will be installed

Gas	Mass (kg)
O_2	4,074,508
H_2O	103,690
N_2	8,555,344
CO_2	7999
Backup Air	50,966,164
CASSSCs	**3211**

Table: Air Composition Masses
Source: Geet J., Microsoft Excel

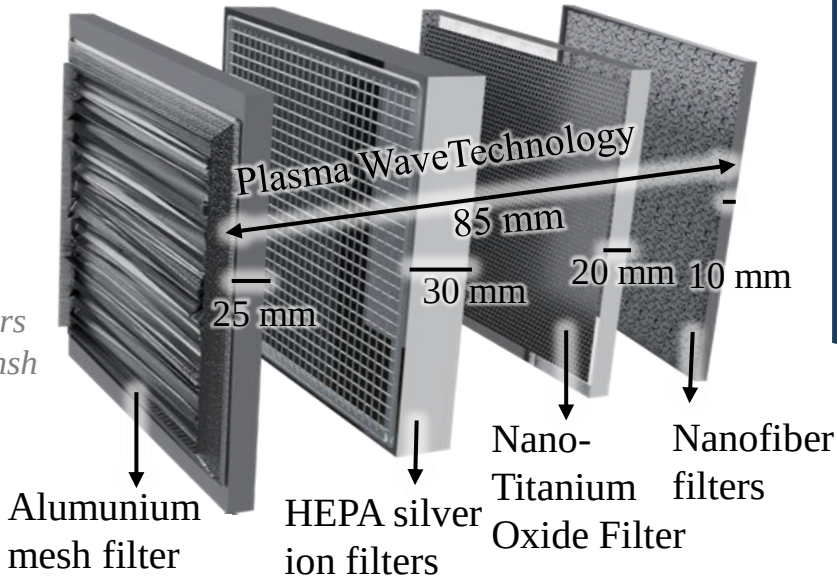


Fig: Air Filters
Source: Devansh M.,Blender

Season	Temperature (°C)	Light Intensity (lux)	Humidity (%)
Summer	24-29	36,000	40
Spring	18 -23	29,000	50
Winter	14 -17	21,000	35

Table: Seasons Conditions Source: Geet J., Microsoft Excel

Different seasons will be simulated with each season lasting for 4 months to simulate a more natural enviornment

Hospitals, clinics, and research centers will include specialized **Polyacrylonitrile Nanofibrous air filters**

FOOD PRODUCTION



Fig: Aeroponic Farm Source: Kelly Z., Blender

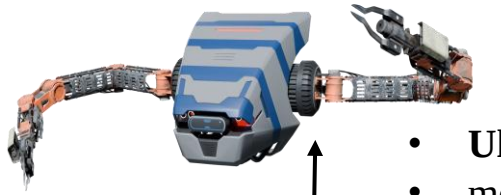


Fig: Cropbot Source:Yatharth G G., Blender

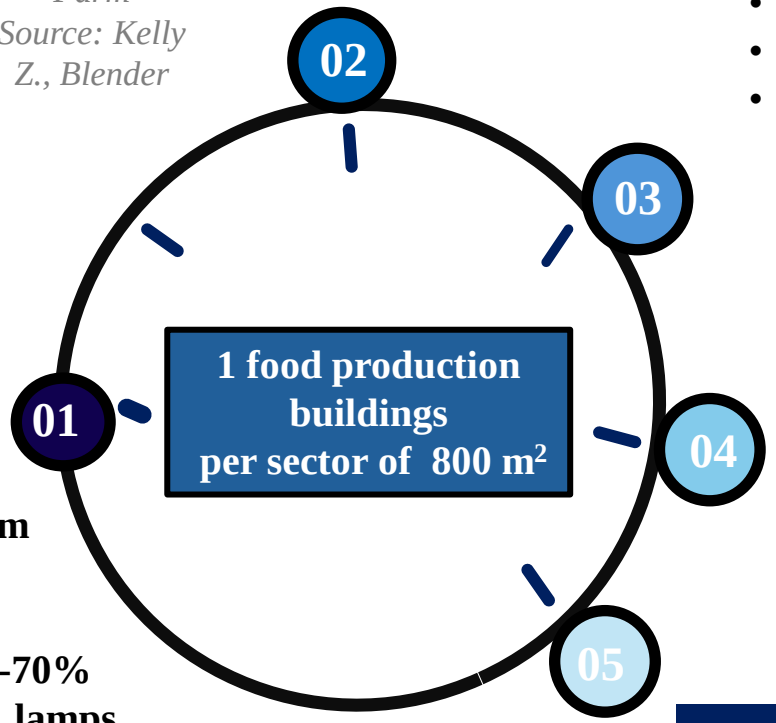
GROWING

- **Ultrasonic Fogger Aeroponic System**
- mostly automated
- staffed with **2 human overlookers**
- Temperature: **18-25˚C**, Humidity: **45-70%**
- **Full Spectrum LED E27 lamps, photosynthetically active radiation**

- **Cropbot** will do harvesting & sowing
- high-pressure ultrasonic mist nozzles precision arms, soft pneumatic actuators

The food processing unit will alter the raw ingredients into bread, spices, curd, etc

FOOD PROCESSING



It will have **11 floors**-
2 floors - packaging & storage
2 floors -food processing
1 floor- milk and meat
6 floors- vertical farming

PACKAGING

- **Packed bed bioreactors** will produce **bacterial cellulose called SCOBY**
- overwrapped with **aluminum foil**
- flushed with **nitrogen 3 times**.
- **10-month** shelf life

STORAGE

- Stored in the **dry storage area**
- Humidity- **15%** Temperature - **18°C**.
- **20%** of the total will go into storage **each month**

SELLING AND DELIVERY

- Delivery vehicles subcontracted to ***Drones and Delivery***
- Sales outlets include cafes, restaurants, and grocery stores
- Residents can order food using Oracle.

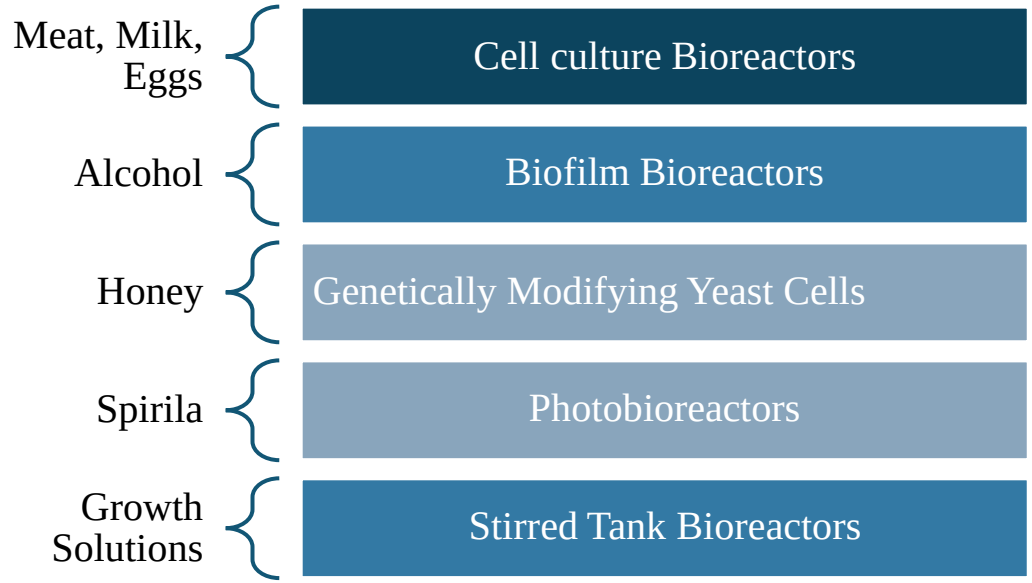
FOOD PRODUCTION

- Delivery vehicles subcontracted to ***Drones and Delivery***
- Sales outlets will include cafes, restaurants, & grocery stores
- Residents would also be able to order food using NeuroVision (personal device)

CONSUMABLE	SOURCE	CASSSCs
Food	Made in Aeroponic Farms	454
Water	***Stuff Of Life***	52
Toiletries	Earth, ***Scrub-A-Dub-Dub,*** and ***Bottom Cleaners***	10
Medical Supplies	Earth	10
Wearables	Earth	13
Kitchen utensils	Aresam	5
Household supplies	***Remotely Local Products***	28

Table : Consumables, Source: Noah., Microsoft PPT

- All food and consumables will be supplied through ***Drone & Delivery***
- Periodic Deliveries to doorsteps



- **Edible Landscaping** will cover **150 m^2 per dome**.
- Soil is subcontracted to ***Soil Solutions***
- ground vehicles for harvesting from ***Drones and Delivery***
- The plant waste produced will be supplied to ***Bottom Cleaners***

WASTE & POWER MANAGEMENT

3.2, 3.4

Agricultural Waste | Recyclable Waste | Non- Recyclable Waste

- Supplied to ***Bottom Cleaners***
- Repurposed (50 % of all waste gets repurposed)
- Transported to Landfill Building

- Recycling of waste will be subcontracted to ***Toss it to me***
- 1112 m2 each **unpressurized, conversion, and landfill building**
- Sewage handling systems and toilets will be subcontracted to ***Waste Products***
- **369 m² 9 toilet paper supply systems** will be subcontracted to ***Bottom Cleaners***
- Ground vehicles for waste collection subcontracted to ***Drones and Delivery***

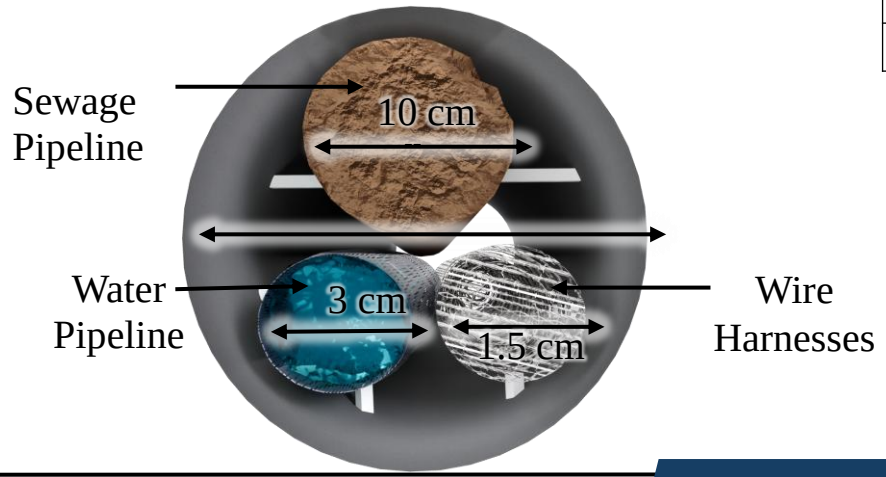


Fig: Cross-Section of Utility Routing Source: Devansh M., Blender

PRIMARY SOURCE:

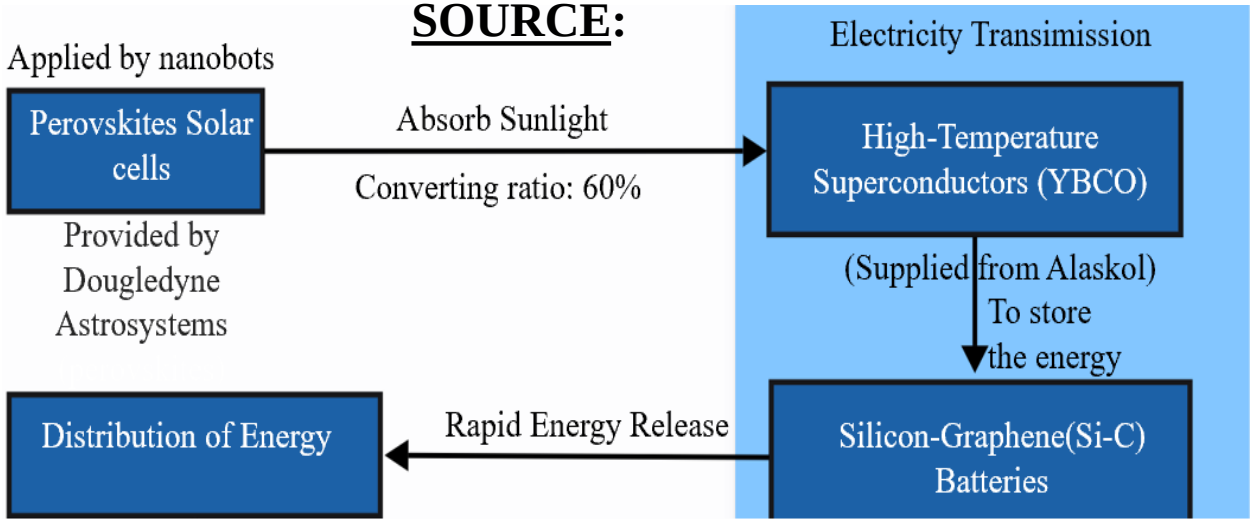


Fig: Power production, Source: Harshit A, Dan Y., Charlie S, Geet J., Powerpoint

Allocation	Power at IOC	Power increase after each pass	Power at OC
Residential	6000 kW	2400 kW	18,000 kW
Food	15,000 kW	5000 kW	40,000 kW
Automations	`750 kW	-	1500 kW
Total Power	**21,750 kW**	**7400 kW**	**59,500 kW**

Fig: Power Allocation, Source: Geet J., Powerpoint

BACKUP SOURCE:

Hydrogen fuel cells will be used as a backup source during contingencies or no sunlight positions. These will be stored in a separate 120 m^2 area with the fuel cells. Water produced will be stored in the safe water supply

WATER MANAGEMENT

FIRE FIGHTING

- Class A and K fires will be detected using smoke detectors and thermal sensors
- Class B & D fires will be detected by infrared sensors
- Class C fires will be sensed by fault detectors
- Inert Gas Fire Suppression System or **IG-541** will be deployed for Class A,B & C fires
- Copper dry powder will be used for Class D fires
- Potassium Acetate mist will be used against Class K fires

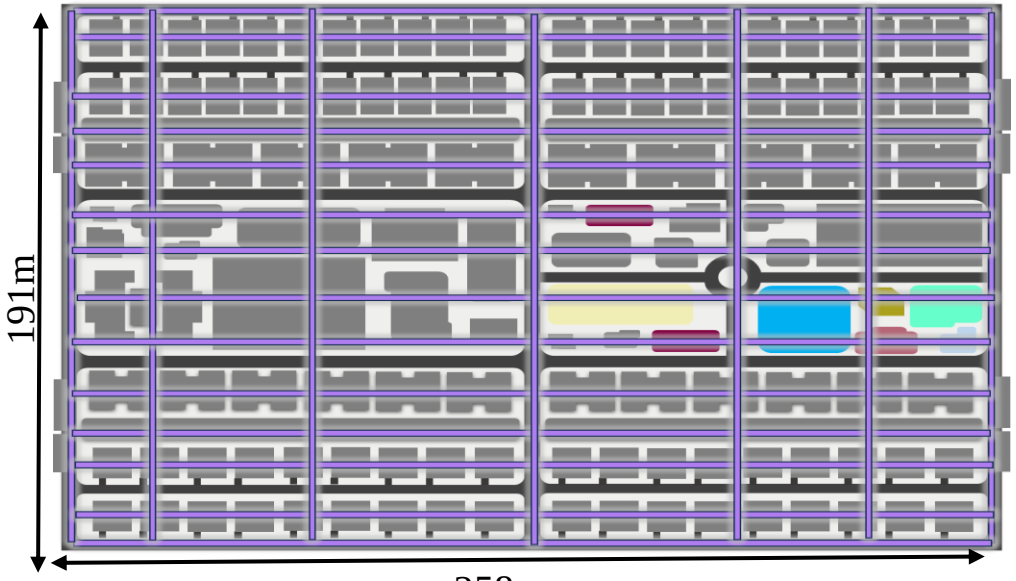


Fig: Sewage Routing
Source: Aatreyi S.,Devansh M., Microsoft PPT.

Purpose	Quantity at IOC (L)	Increase each Pass	Quantity at OC
Residential	150,000	60,000	450,000
Recreational	3,069,845	1,008,865	6,281,226
Food	442	42	1326
Total Water	3,220,287	1068907	6732552
Backup Water	9660861	3206721	20197656
CASSSCs	70	24	110

Table: 3.9: Water Consumption per year Source: Geet J., Microsoft Excel

Subcontracted to ***Stuff Of Life,*** recycled by ***Clean Up Your Act***

800 **m²** cylindrical water supply per sector made from **repurposing CASSSCs** by the ECB

A **Safe Water Supply** present in each dome will provide water up to **3 days** in case of contingencies

Backup water for 3 days in case of contingencies

Air Tank

Water Tank

Conversion Building

Aeroponic Facility

Horizon Stations

Bottom Cleaners

Landfill Building

PERSONAL DEVICE

NeuroVision

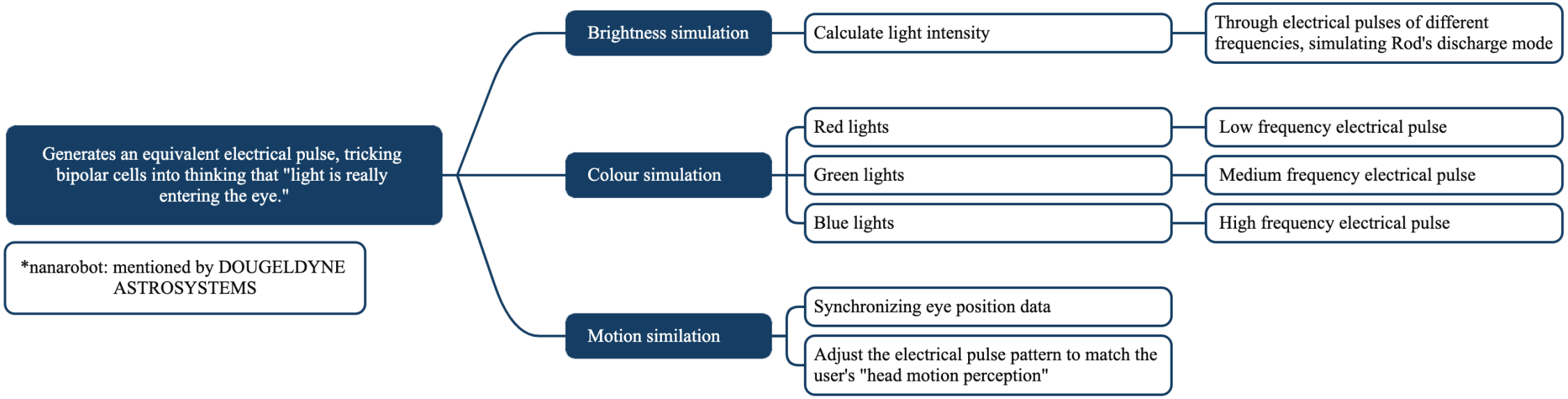


Table: NeuroVision Flowchart; Source: Selina W., Xmind

Bioelectronic device designed for visual communication and neural manipulation using nanobots.	Genetically engineered organisms transmits electrical signals in the eye to **replicate** the actions of **rod and cone cells**.	Nanobots will be **reprogrammed** to interpret **radio wave signals** which will be converted into **chemical signals** that the micro-organisms receive.	**Enables two-way communication**, allowing images, videos, or data to be directly projected into the user's field of vision.	Overlays **augmented reality (AR)**, manipulates visual environments, making it beneficial for people with poor eyesight or blindness.

ROCKDONNELL will supply the genetically engineered organisms via *LitigationLimiters*. **DOUGELDYNE ASTROSYSTEMS** will supply the nanobots via *LitigationLimiters*.

Residential automation will be subcontracted to *Bots4U* to handle tasks requiring **manual labor**.

PERSONAL DEVICE

The space settlement will feature **17 servers** dedicated to different tasks. They will feature **Neural Mesh Storage** provided by **ROCKDONNEL** via *litigation limiters* for organization and capacity. **AES 512-bit** and **SHA-512** will be used in encryption for data transmission to ensure storage integrity.

QTY	FUNCTIONS	SPECS
3	**Main Control** Stores user data security logs, sensor data	**Processor:** RISC-V, 21 EFLOPS **Memory:** 4.5 EB **Storage:** 25 EB
5	**Application** Manages robots & automations	**Processor:** RISC-V, 10.5 EFLOPS **Memory:** 2.1 EB **Storage:** 1 EB
5	**Communication** Internal/external communication	**Processor:** RISC-V, 5.5 EFLOPS **Memory:** 4.5 EB **Storage:** 2.5 EB
3	**Miscellaneous** Backup for critical systems	**Processor:** RISC-V, 3 EFLOPS **Memory:** 0.5 EB **Storage:** 0.5 EB
1	**Backup** Backup for Logs and data	**Processor:** RISC-V, 2.5 EFLOPS **Memory:** 4.5 EB **Storage:** 25 EB

Table: Server Specifications, Source: Yatharth G., PPT



Fig: Echo, Source: Selina W., Blender

Echo

Uses **bone conduction** to aid people with hearing disabilities.

Features an **ARM** processor for **AI functions and general tasks**

Contains a wireless SoC and antenna for connectivity

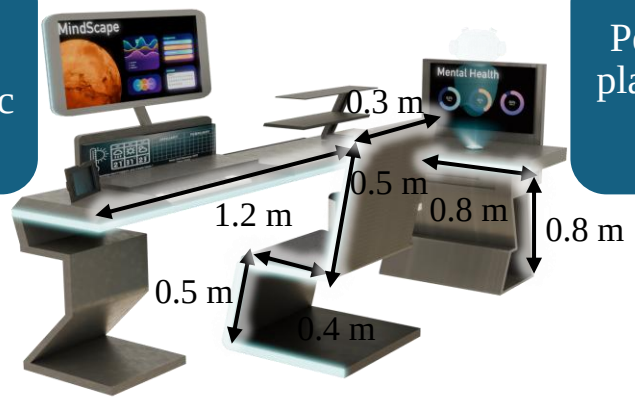
Additionally features **voice and touch control** for commands

LIVEABILITY

MindScape

Accesses a vast digital library.

Contains capacitive touch sensors with haptic feedback.



Personalizes task plans and manages workloads.

Links with NeuroVision for real time feedback.

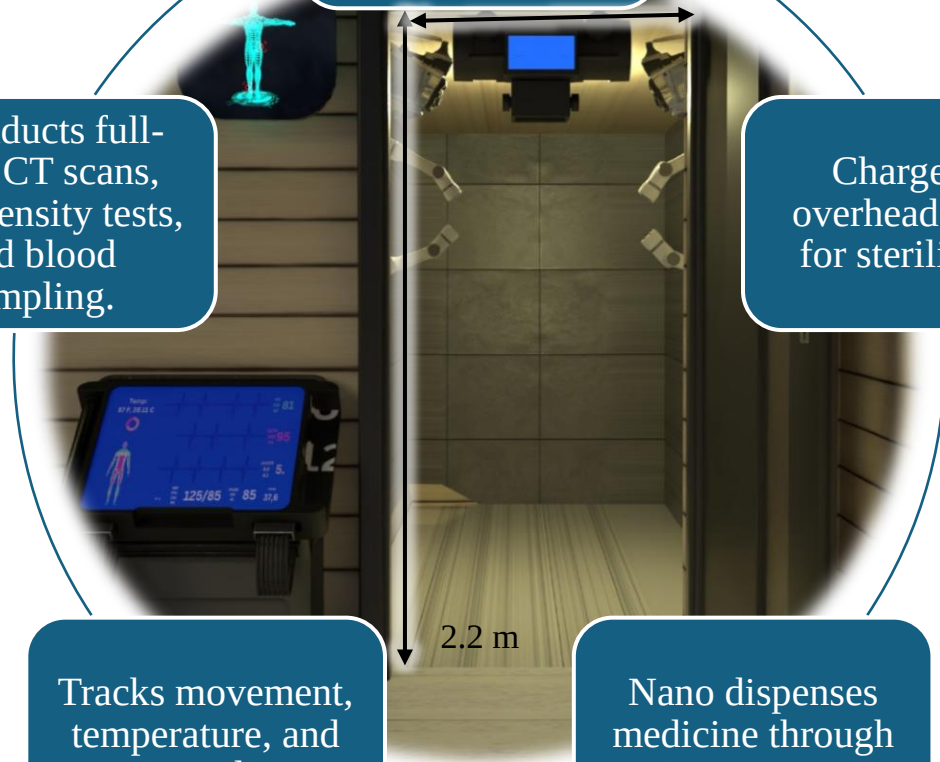
Gives personalized advice for children with special needs.

Fig: MindScape, Source: Suryansh P., Blender

Halo-Scan

A compact health station tracking vital signs.

It conducts full-body CT scans, bone density tests, and blood sampling.



Chargeable overhead device for sterilization

Tracks movement, temperature, and air quality

Nano dispenses medicine through mists or sprays

Fig: MindScape, Source: Karman B., Blender

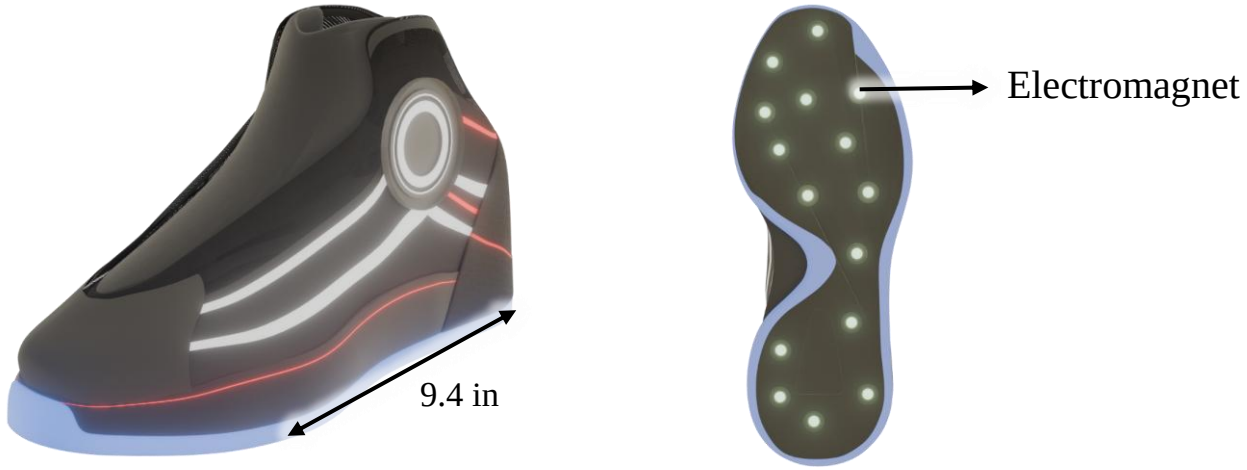
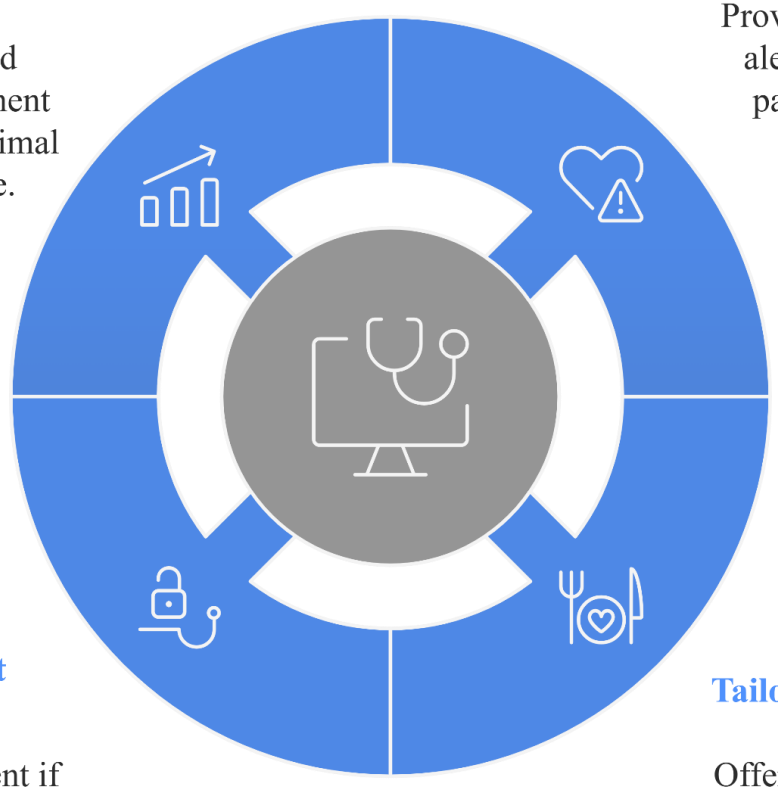


Fig: Moonwalker, Source: Suryansh P., Blender

Moonwalker

- NeuroVision Integration
- Blood oxygen and Respiratory rate sensors
- Vital Alerts
- DXA Sensors
- Electromagnets for Gravity Simulation

HitFit Gym



Performance Tracking

Monitors and adjusts equipment settings for optimal performance.

Alerts for Safety

Provides real-time alerts to ensure patient safety.

Equipment Locking

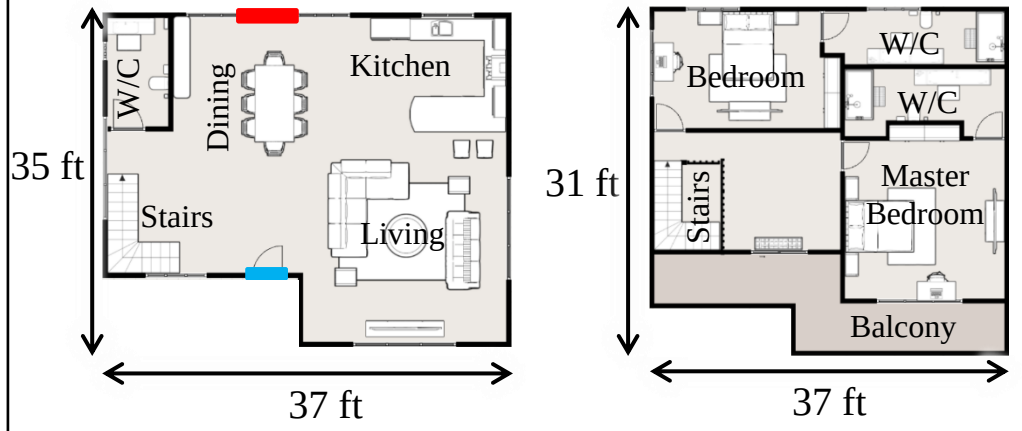
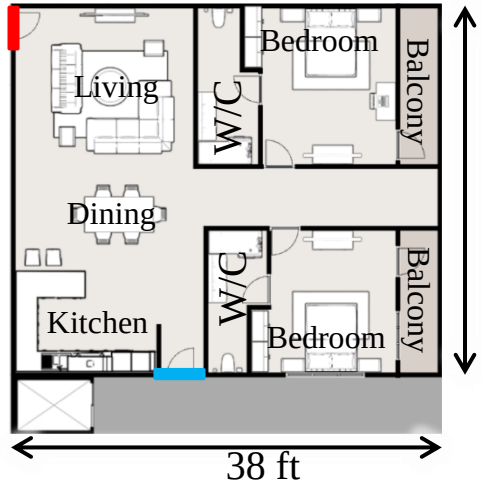
Locks equipment if vitals are unstable to prevent misuse.

Tailored Diet Plans

Offers personalized diet and health plans for users.

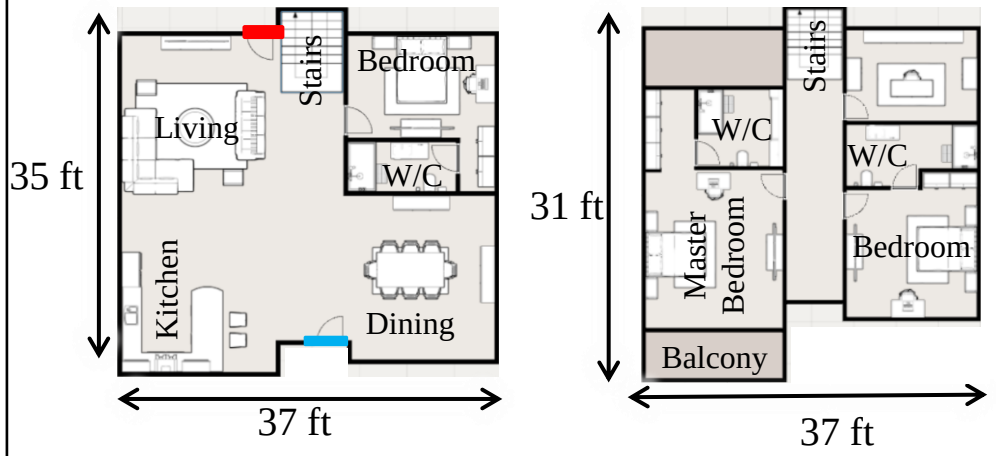
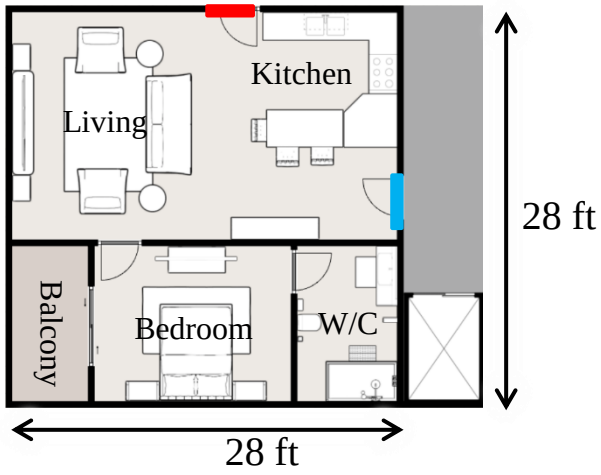
RESIDENTIAL DESIGNS

Halo-Scan (within doorways) will conduct essential medical tests, including bone density assessments, to continuously monitor your health while residing in a low-gravity environment.

EXTERIOR	INTERIOR	Specifications
 *Fig: 2 Bedroom Individual House* *Source: Karman B., Blender*	 *Fig: First & Ground floorplans | Source: Karman B., floorplanner.com*	Individual house 2 Bedroom Permanent Residents 1351 sq. ft. 44 Houses
 *Fig: 2 Bedroom Apartment* *Source: Karman B., Blender*	 *Fig: First & Ground floorplans | Source: Karman B., floorplanner.com*	**Apartment** 2 Bedroom Permanent and Transient Residents 1738 sq. ft. 10 Apartment buildings

Entry Exit

RESIDENTIAL DESIGNS

EXTERIOR	INTERIOR	Specifications
 *Fig: 3 Bedroom Individual House* *Source: Karman B., Blender*	 *Fig: First & Ground floorplans | Source: Karman B., floorplanner.com*	Individual house 3 Bedroom Permanent and Transient Residents 1480 sq. ft. 36 Houses
 *Fig: 1 Bedroom Apartment* *Source: Karman B., Blender*	 *Fig: First & Ground floorplans | Source: Karman B., floorplanner.com*	**Apartment** 2 Bedroom Transient Residents 841sq. ft. 12 Apartment buildings

Entry Exit

Education & Curriculum:

CURRICULUM SUMMARY:

1. About being on board



Information on spacesuits	Evacuation routes	How to use security & life devices (e.g. in low-g places)

Category	Description	Quantity
Instructors	Teachers	50 people
	Instructors	30 people
Rooms	Classrooms	50 rooms
	Simulation rooms	30 rooms
Staff	Cleaners	16 people
	Maintenance Men	8 people

Table: Number Of Instructors and Rooms
Source: Ahmad Ali A., Microsoft PowerPoint

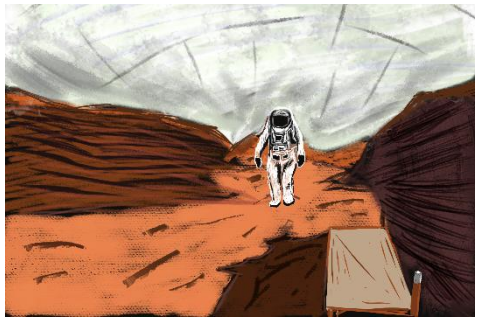
2.About being **off board** (on Mars)

- Basic Mars knowledge:

1. Temperature
2. Atmosphere composition
3. Possible disasters
4. Different weathers
5. Day & night length (in different seasons)

- Daily life:

1. Transportation: How to walk on Mars, How to use vehicles
2. How to use life maintenance systems (e.g. spacesuits, oxygen system)
3. Health & Others
4. Basic medical knowledge (how to deal with possible sickness on Mars)
5. Exercises for keeping healthy
6. How to repair common equipment on Mars
7. Mental Resilience Facilities





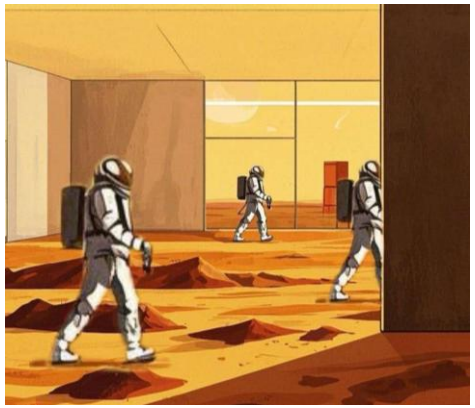


Fig: Hands-on training
Source: Sara X., Notein

INTERNAL TRANSPORTATION

Mechanism Of Transport in Microgravity

The vehicle's wheels equipped with controllable electromagnets rotate as it moves

When a section of the wheel makes contact with the ferrous embedded road, pressure sensors detect the contact point

The electromagnets only activate at the contact points, creating a temporary magnetic grip that enhances traction

As the wheel continue rotating, previous contact points deactivate, ensuring only the active contact area engages with the road, minimizing energy usage and optimizing movement

1. LINEAR SYNCHRONOUS LIFTS

Uses electromagnetic force for motion

Facilitate transport between the rings

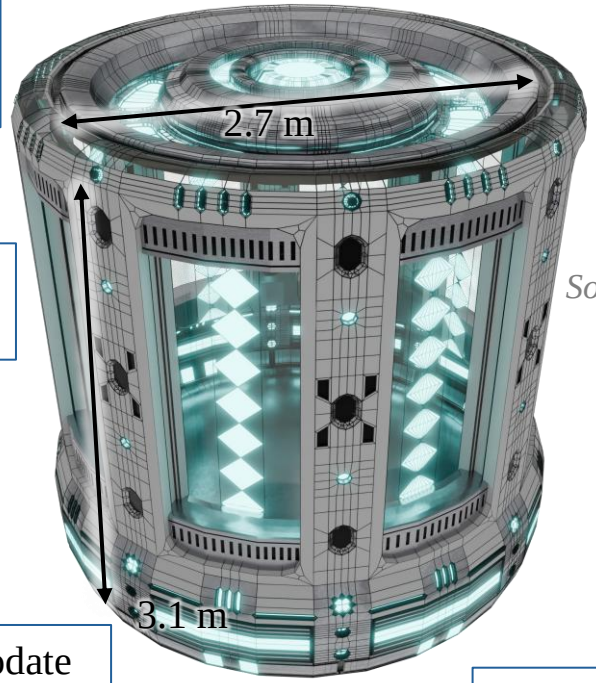


Fig: Lift
Source: Mahum K., Nakshatra M.S., Blender

accommodate **50 passengers**

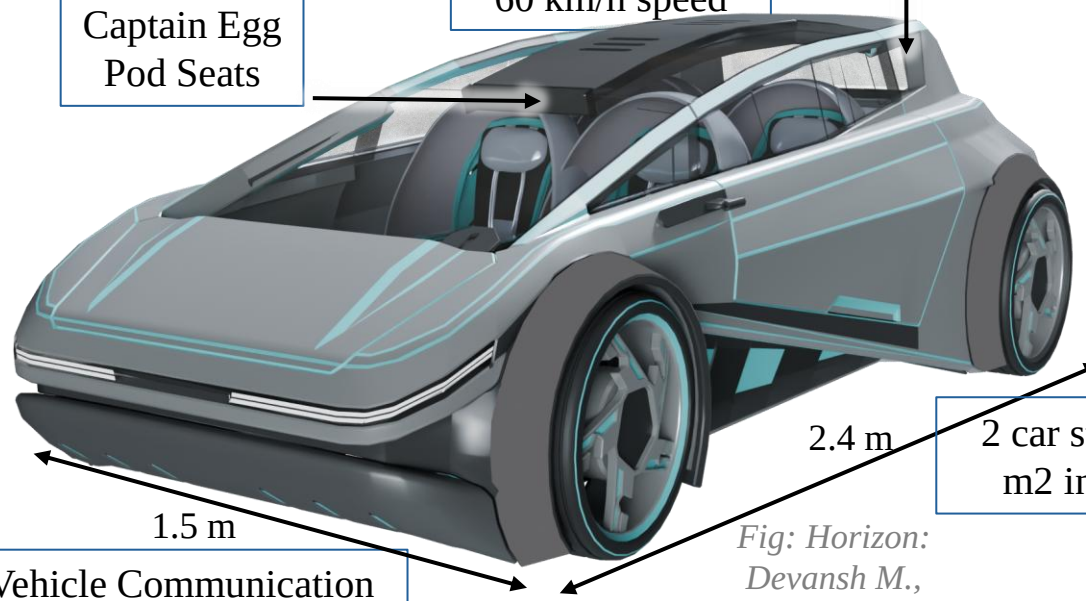
2.HORIZON

4- seater autonomous electric pod

AR Windows for information & entertainment

60 km/h speed

Captain Egg Pod Seats



2 car stations of 320 m2 in each sector

Cell To Everything Vehicle Communication

Fig: Horizon: Devansh M., Blender

INTERNAL TRANSPORTATION

3. AMBULANCE

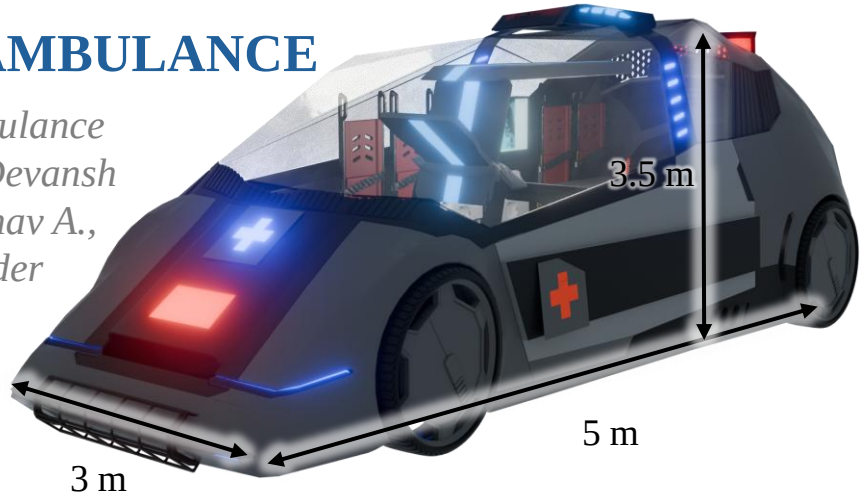


Fig:Ambulance
Source: Devansh M.,Krishav A., Blender

AI biometric systems with automated 3D body scanner

Accommodates a 5 - people medical team

Category	Type
Cameras	Narrow Forward Camera
	Wide Forward Camera
	Forward Looking Side Camera
	Rear View Camera
	Rearward Looking Side Cameras
Sensors	Lidar
	Ultrasonic Sensors
	IMUs
	GPS

Table: Cameras & sensors of Spectre & Ambulance
Source: Geet J., Microsoft Excel

Vehicles	OC Quantity	Increase After each pass	IOC Quantity
Spectre	440	172	1300
Ambulance	5	3	20
Lift	4	-	4
CASSSCs	**40**	**20**	**140**

Table: Quantity of Vehicles
Source: Geet J., Excel

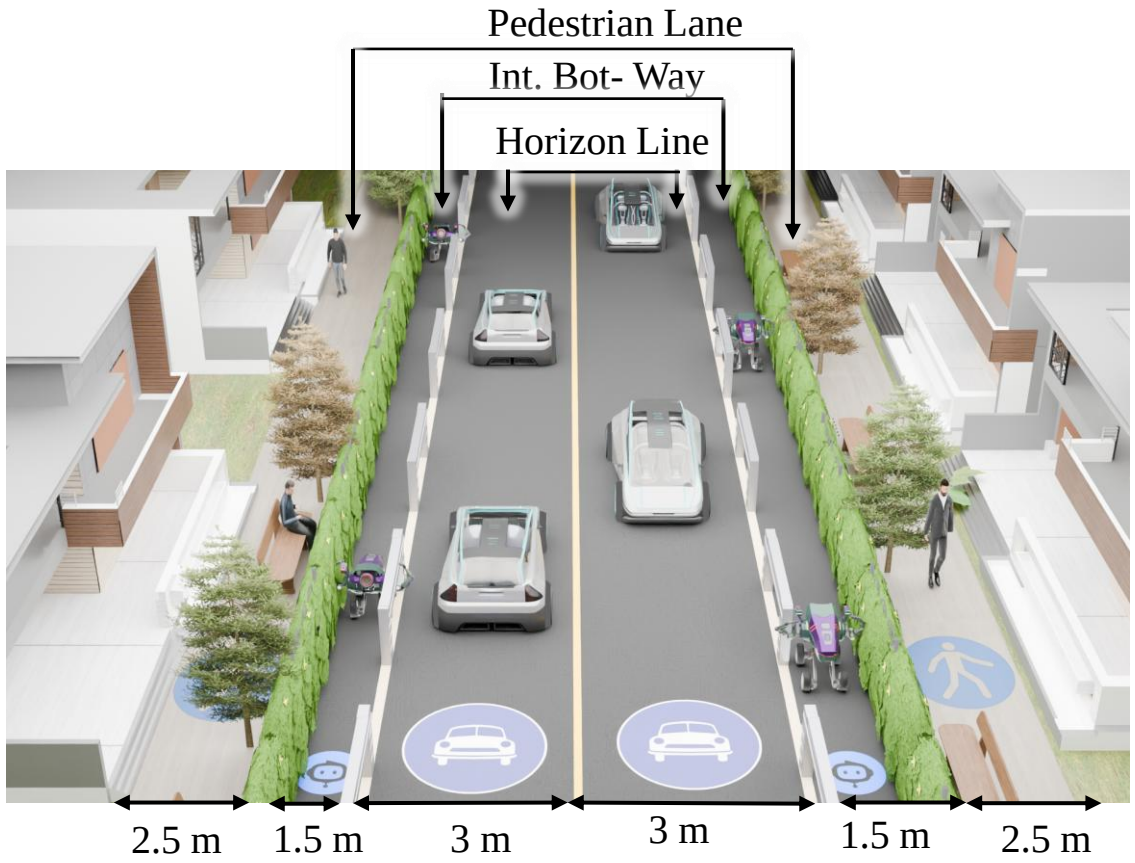


Fig: Road Layout Source: Devansh M., Blender

SAFETY FEATURES

HUMAN INTERVENTION

Software and technology **glitches** in devices & guest automations which occur frequently

Radiation leak, Solar flares, Conflagration, **Major defects** in the structure

Resolving device glitches and technological **malfunctions**

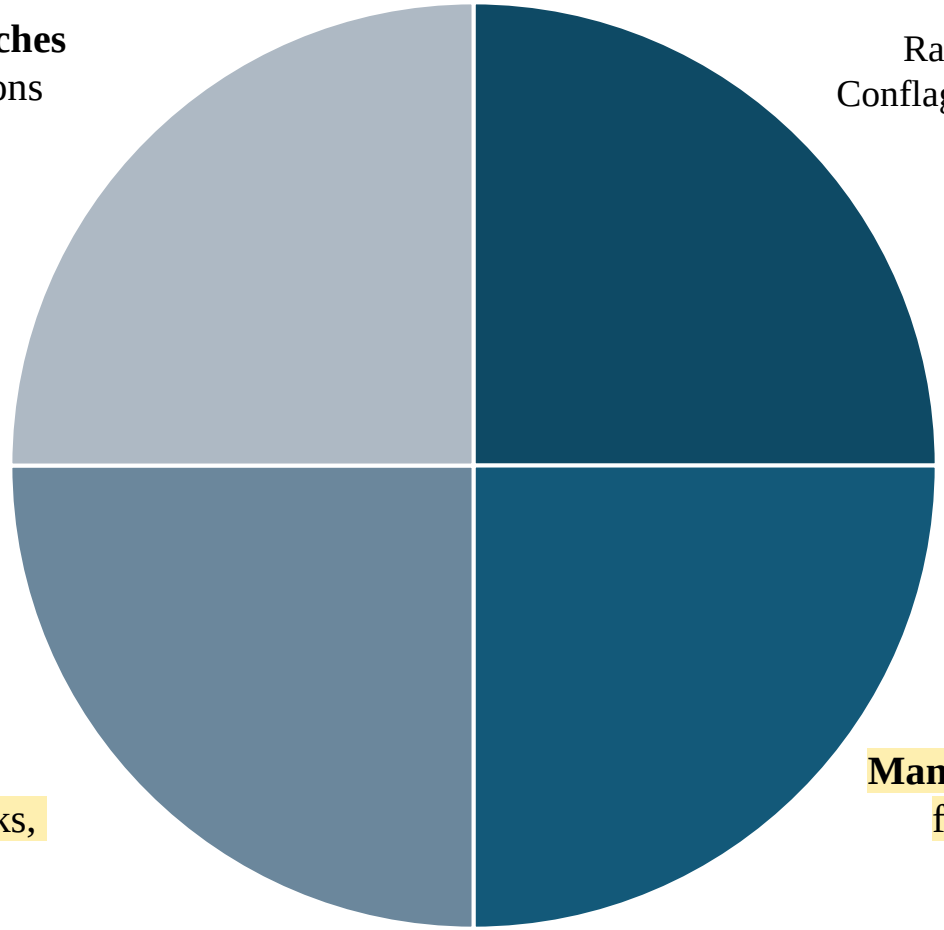
Medical **emergencies & Evacuations**

Routine updates, Minor **maintenance** issue

Equipment failure, Authorization failure Power & communication failure

Battery changes, **inspections**, visual checks, and user support

Manual override for authentication failures and malfunctioning systems



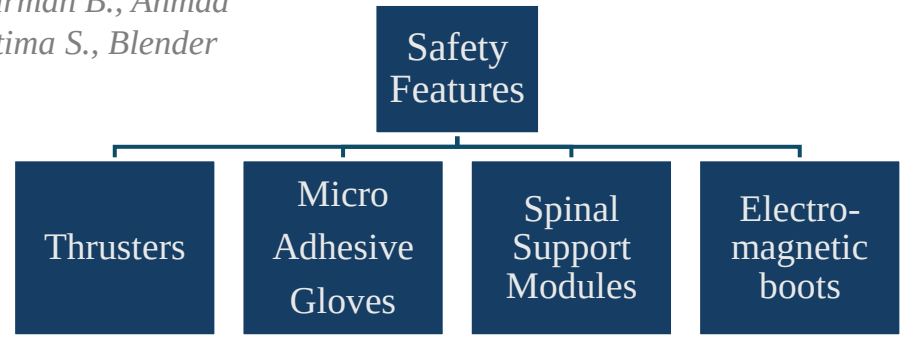
SPACESUITS & SAFETY

In case of severe solar flare and radiation events:

- In case of emergencies, portable shelters subcontracted to ***Extreme Survival Technologies (EST)*** will be deployed to convert the sports facility into shelters.
- Houses will be constructed with materials which will provide radiation and solar flare protection.
- The undergarment layer of the spacesuits will include a layer of Polybenzoaxine for additional radiation protection.



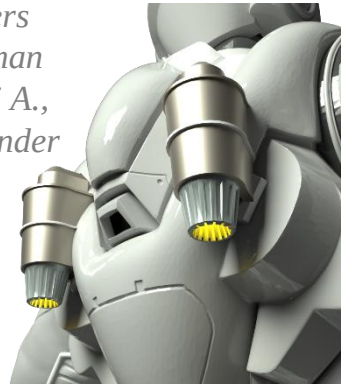
Fig: Spinal Support Module
Source: Karman B., Ahmad Ali A., Fatima S., Blender



The **MSV Suit:** a highly modular and adaptable spacesuit designed for **seamless transition between IVA and EVA operations.** Quantity: 10,500 units.

EVA Adaptability: micrometeoroid-resistant Kevlar layers and **ceramic plating** can be attached for full EVA capability, with **flexible silicon buckystructure fibres.**

Fig: Thrusters
Source: Karman B., Ahmad Ali A., Fatima S., Blender



EVA Adaptability: micrometeoroid-resistant Kevlar layers and **ceramic plating** can be attached for full EVA capability, with **flexible silicon buckystructure fibres.**

The **base pressure garment,** which can be worn normally will consist of a **modular design**, providing **4.3 psi** of pressurization.

Articulated neck joints and **spinal support modules** of the base pressure garment will offer **flexibility** and allow seamless attachment of essential modules like Oxygen, Nitrogen (propulsion), etc.

Table: MSV Spacesuit Layers
Source: Ahmad Ali A., Microsoft PowerPoint

LAYER	MATERIAL
Outer Layer	Nylon
Temperature Control	Mylar insulation
Pressure Layer	Neoprene



Fig: Micro-adhesive gloves
Source: Karman B., Ahmad A., Blender

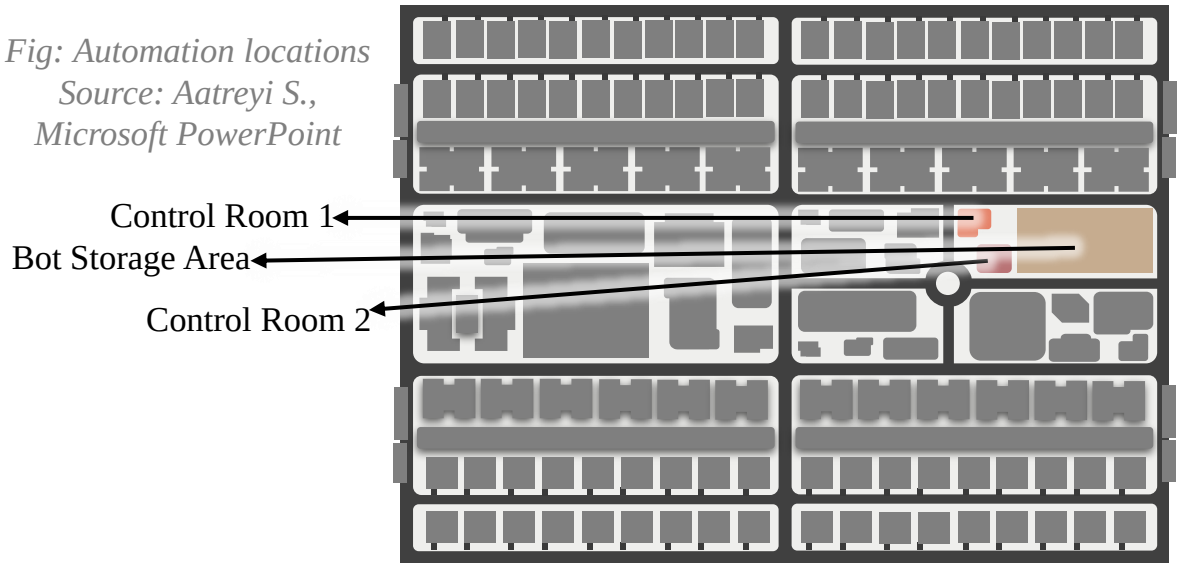
MAINTENANCE AND REPAIR



Fig: Control Center, Source: Allowyn H, Stuti P, Blender

ECBs and ICBs will be repurposed for **maintenance** and **repair**	**Fiber-Bragg Gratings (FBG) will be used** for high-precision measurements of critical structural components	**Human interaction** will be used in **healthcare, control**, and **psychological support**.

Fig: Automation locations
Source: Aatreyi S., Microsoft PowerPoint



Bot/Device	Quantity
ECB	2010
ICB	1452
NeuroVision	9500
Echo	9500
Halo-Scan	1772
Mindscape	1772
Moonwalker	9500
HitFit Gym	4

Table: Bot and Device Numbers, Source: Stuti P, Manan B., PPT

"**Micro-abacus**" modular computers will be subcontracted for servers from **ROCKDONNELL** via *Litigation Limiters*. Advanced **cooling system** is subcontracted to *CoolWave Dynamics* for **thermal management**

CONTINGENCIES

CONTINGENCY	DETECTION	SHORT TERM FIX	LONG TERM FIX
Cosmic Radiation	Radiation sensors	Equip **IVA suits** provided to the residents	Repair external shielding using **ECB**
Conflagration	**Class A & K:** Smoke detectors and thermal sensors **Class B & D:** Infrared sensors **Class C:** Arc fault detectors	Automated emergency systems deployed depending on the **class of fire,** like the Inert gas fire suppression (**IG-541**) system installed, **dry powder extinguishers**, and **wet chemical extinguishers**; Evacuation routes shown by **photoluminescent strips** on route	Replace and reinforce interior and structural materials with **Intumescent coatings** as a flame-retardant option; Use heat-resistant wiring.
Construction bot power failure	Error logs in the control room; shut down of bot.	Switch to **backup power source**; Deploy **backup bots** or take **manual control** over malfunctioning units	Change **bot power systems**; **Replace broken parts** using ECBs; Upgrade software to prevent malfunctions.
Life support failure	**SCADA** system, Pressure & oxygen sensors	Activate **emergency oxygen** and isolate malfunctioning sections	Enhance life support with **redundant** oxygen, water recycling, and air filtration **backups.**
Solar flare	**MAVEN's** solar flare detection technology and **Solar Dynamics Observatory (SDO: Earth)**	Equip **IVA suits** provided to the residents	**Deploy ICBs** to clear residue
Authorization failure	Multiple failed attempts of authentication	**Manual override** by human personnel	**Replace biometric** systems
Water contamination	pH sensors	Switch to **backup water** supplies.	**Replace filtration** and purification systems
Plant growth failure	**Environmental sensors** (light, temperature, moisture, pH)	Switch to **backup food** reserves	Adjust **environmental controls** for water supply and nutrient infusion to restore balance.

Table: Contingencies; Source: Stuti P., PPT

COMMUNICATION

EXTERNAL COMMUNICATION – Near Transmission Lasar Network

- Transmitter and receiver Infrastructure on Aresam (Pheibos), Deimos, Alexandriat, Benevectoras (central hub), and Earth.
- System calculates orbital positions to select shortest laser path.
- Alternate routes are chosen if Sun blocks direct transmission.
- Laser Optics will be used; Fiber Optics subcontracted to ***Zap Industries!***
- Antenna subcontracted to ***OrbitLink Communications.***
- Uses **AES-256** encryption.

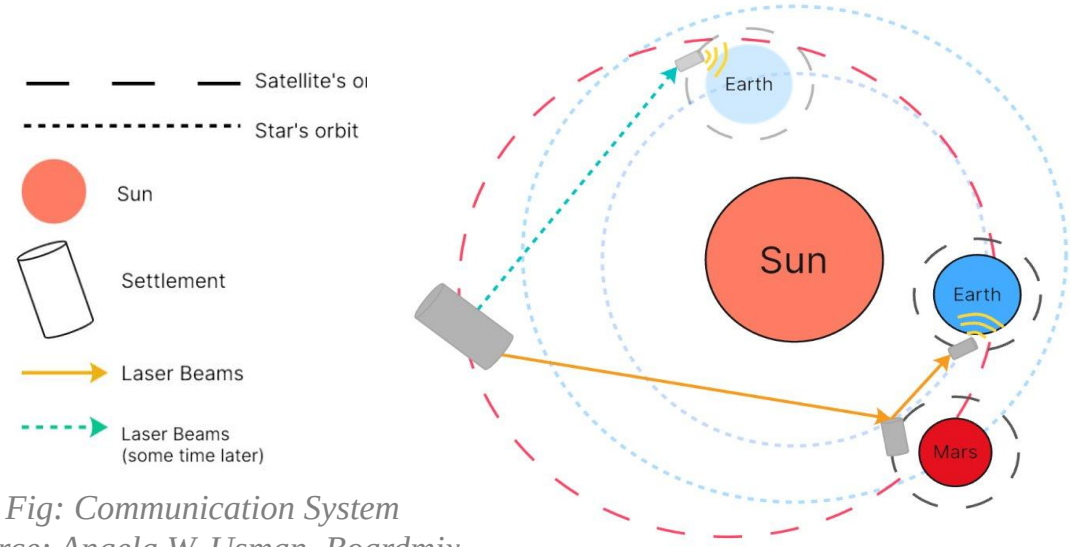


Fig: Communication System
Source: Angela W.,Usman, Boardmix

INTERNAL COMMUNCATION

Uses Terahertz waves - 300Tbps speed and a 14g network

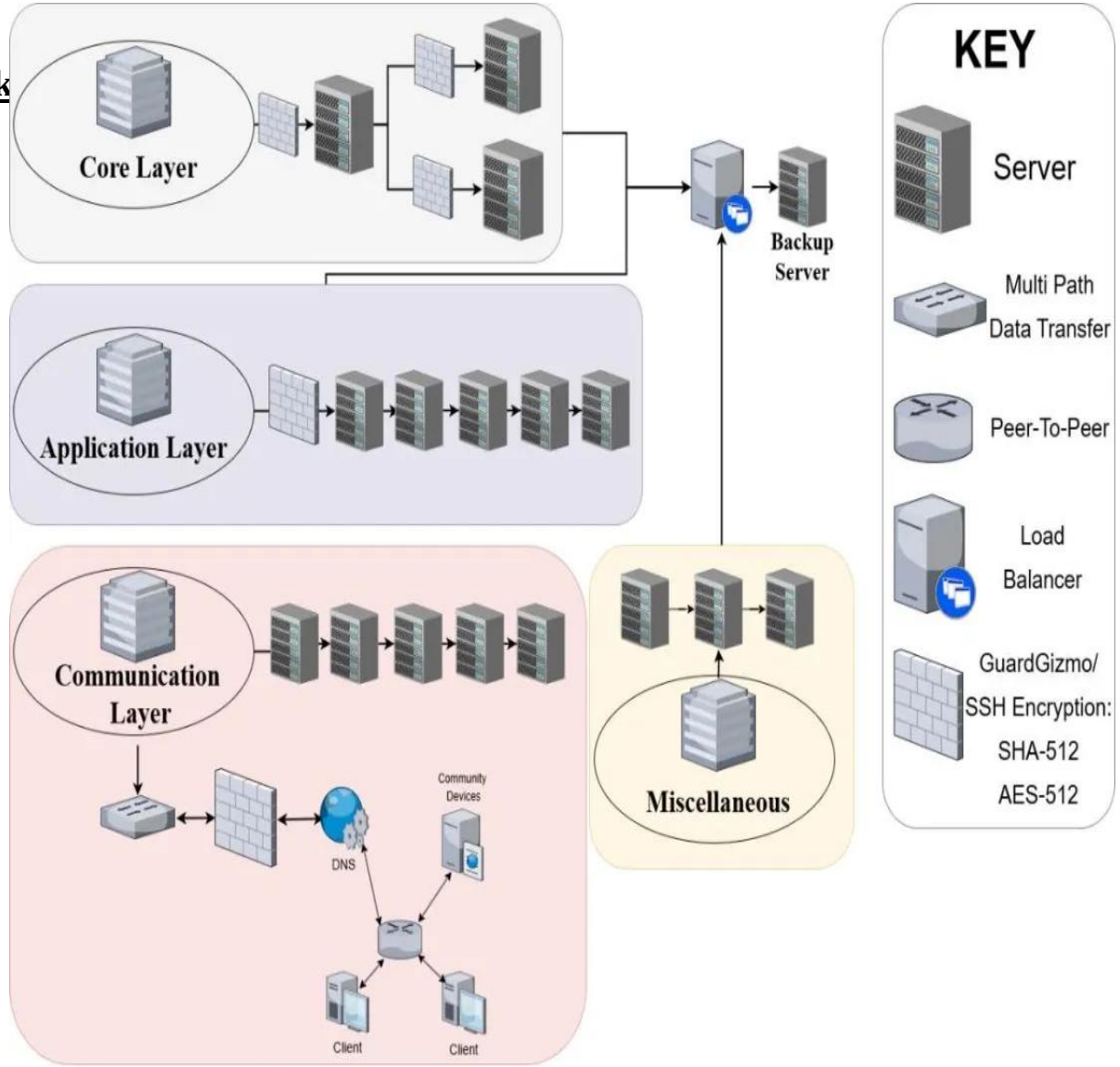


Fig: Network Diagram, Source: Yatharth G G., PowerPoint

IDENTIFICATION

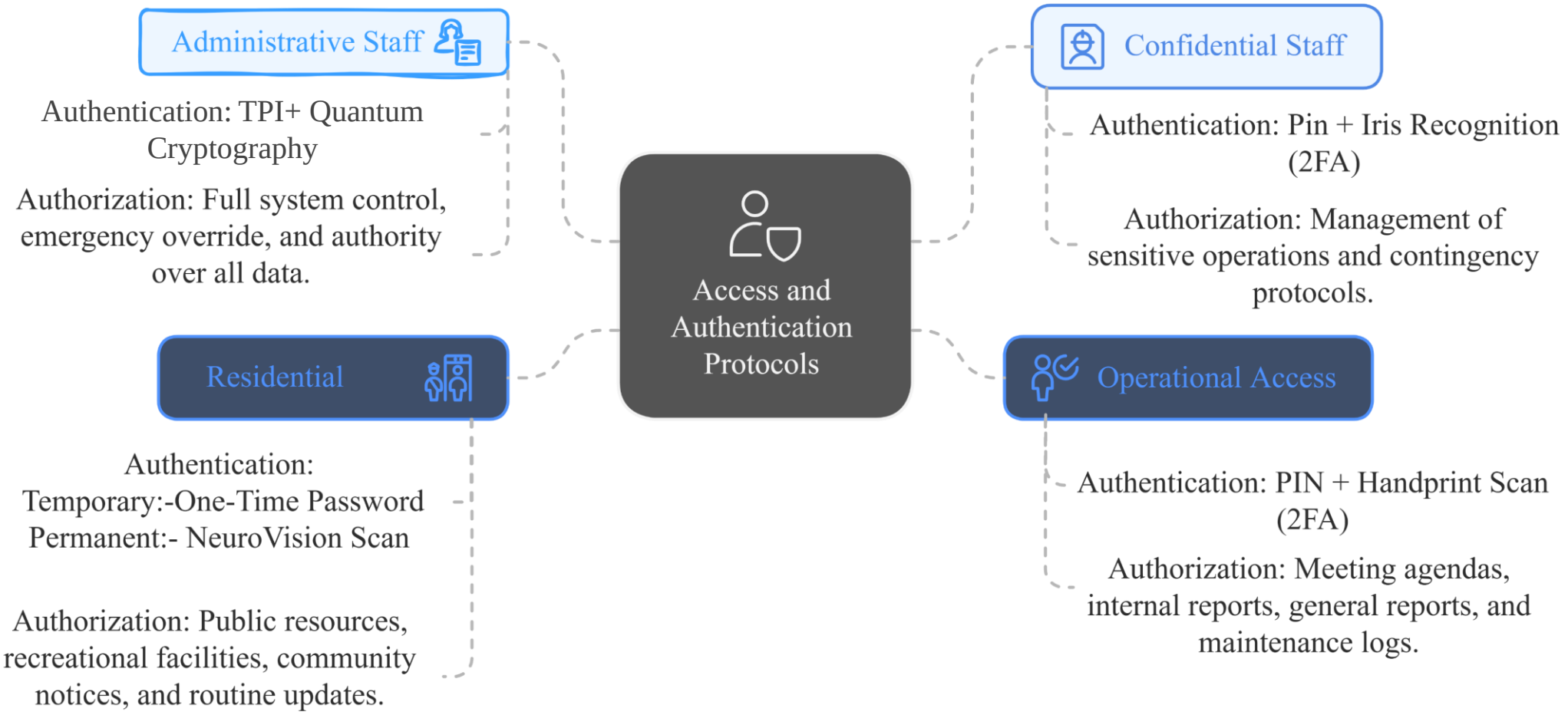


Table: Identification, Source: Manan B, Stuti P., Xmind

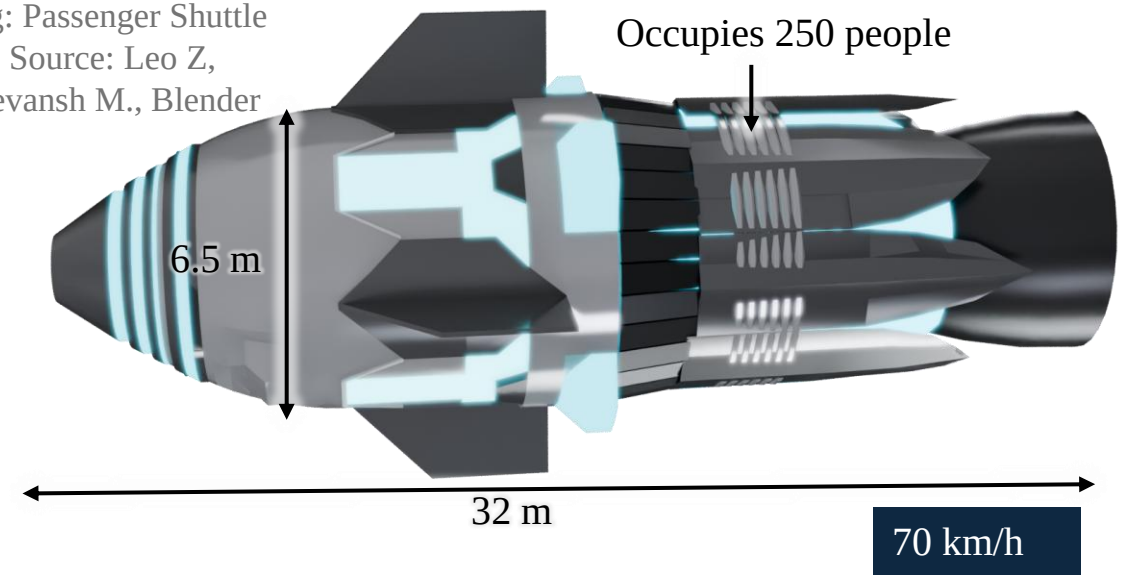
For identification we will be using the **"Zero Trust Architecture" involving Role-Based Access Control (RBAC)** for authentication and security, along with **quantum cryptography** to safeguard sensitive data, which will be subcontracted to *GuardGizmo* for managing all authorization and access protocols.

EXTERNAL TRANSPORTATION

3.2, 3.4

1. PASSENGER SHUTTLE

Fig: Passenger Shuttle
Source: Leo Z, Devansh M., Blender



- 3 ferries will be based at Mars and 3 at Earth
- During Earth or Mars passes, they will run **3 times** a day

Spacious with life support system .

Standardised cargo units for easy transfer.

2. CARGO SHUTTLE

A) Front View

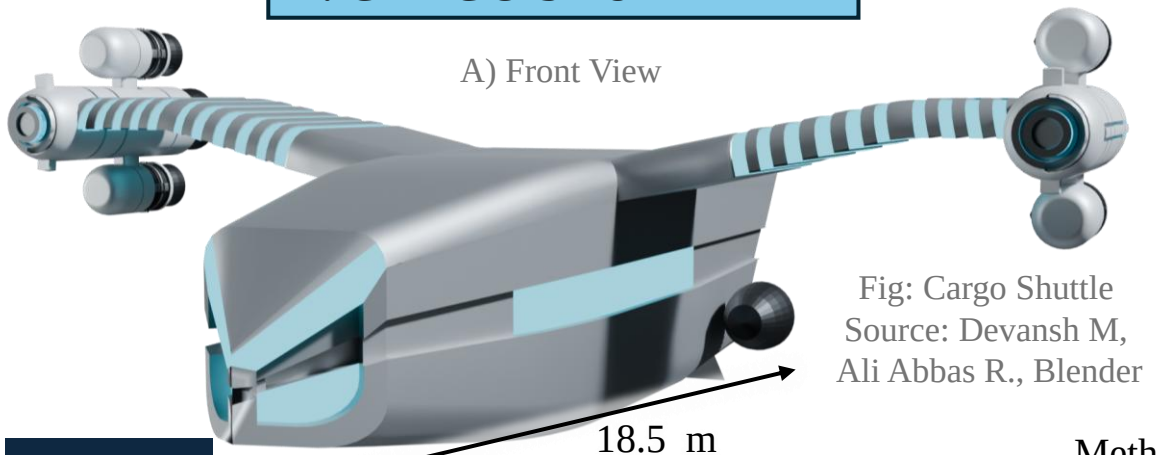
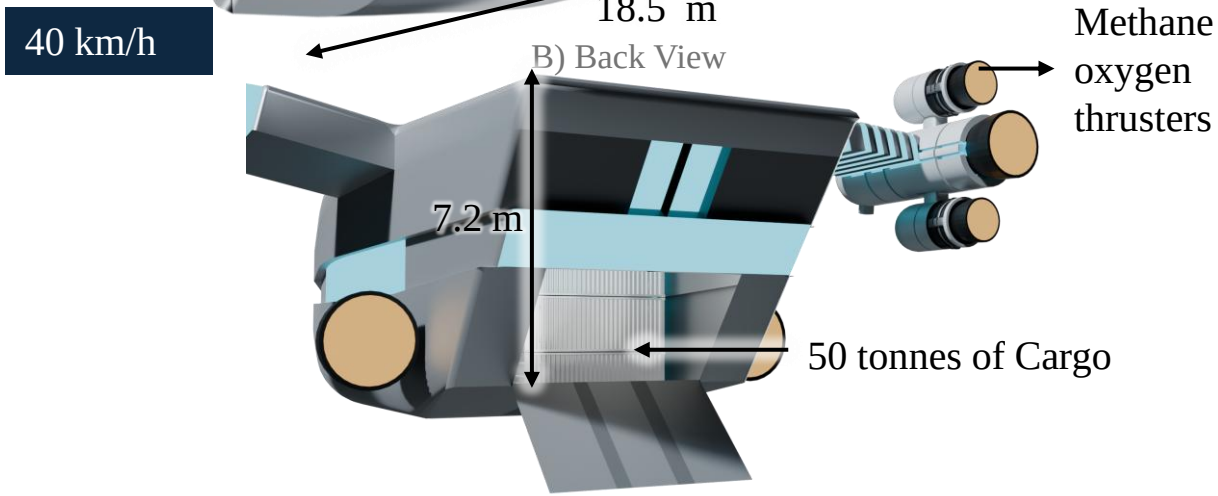


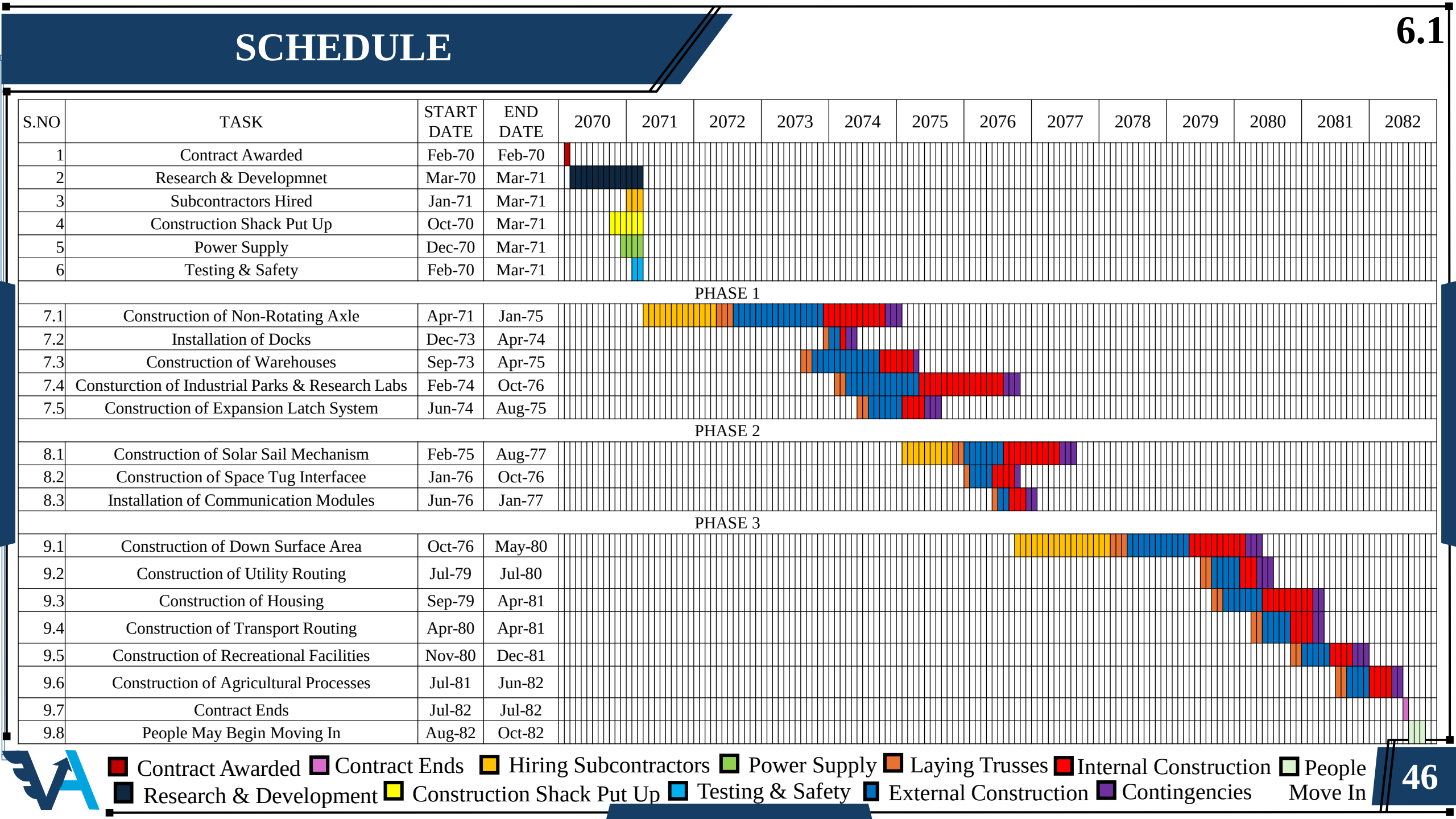
Fig: Cargo Shuttle
Source: Devansh M, Ali Abbas R., Blender



Methane Oxygen Thrusters will be used, methane will be provided by ***Toss It To Me.***

SCHEDULE

S.NO	TASK	START DATE	END DATE
1	Contract Awarded	Feb-70	Feb-70
2	Research & Developmnet	Mar-70	Mar-71
3	Subcontractors Hired	Jan-71	Mar-71
4	Construction Shack Put Up	Oct-70	Mar-71
5	Power Supply	Dec-70	Mar-71
6	Testing & Safety	Feb-70	Mar-71
	PHASE 1		
7.1	Construction of Non-Rotating Axle	Apr-71	Jan-75
7.2	Installation of Docks	Dec-73	Apr-74
7.3	Construction of Warehouses	Sep-73	Apr-75
7.4	Consturction of Industrial Parks & Research Labs	Feb-74	Oct-76
7.5	Construction of Expansion Latch System	Jun-74	Aug-75
	PHASE 2		
8.1	Construction of Solar Sail Mechanism	Feb-75	Aug-77
8.2	Construction of Space Tug Interfacee	Jan-76	Oct-76
8.3	Installation of Communication Modules	Jun-76	Jan-77
	PHASE 3		
9.1	Construction of Down Surface Area	Oct-76	May-80
9.2	Construction of Utility Routing	Jul-79	Jul-80
9.3	Construction of Housing	Sep-79	Apr-81
9.4	Construction of Transport Routing	Apr-80	Apr-81
9.5	Construction of Recreational Facilities	Nov-80	Dec-81
9.6	Construction of Agricultural Processes	Jul-81	Jun-82
9.7	Contract Ends	Jul-82	Jul-82
9.8	People May Begin Moving In	Aug-82	Oct-82



COST

MATERIALS COST PER PHASE

	PHASE 1		PHASE 2		PHASE 3	
NAME	**QTY (kg)**	**COST**	**QTY (kg)**	**COST**	**QTY (kg)**	**COST**
Lunar Regolith Bricks	68167680	$6,407,761,920	–	–	68167680	$6,407,761,920
Aluminum 6061	24345600	$2,239,795,200	–	–	24345600	$2,239,795,200
Electrodynamic Dust Shield	52586046	$6,888,772,026	–	–	–	–
Basalt Fiber	–	–	–	–	90353874	$6,053,709,558
Al 6061	48594584	$3,255,837,128	–	–	48594584	$3,255,837,128
Mg Al Spinel	558034	$37,388,278	–	–	558034	$37,388,278
Fused Silica	42089487	$2,819,995,629	–	–	–	–
Polycarbonate	–	–	–	–	42038423	$3,825,496,493
Electrochromic Windows	28737459	$2,845,008,441	–	–	28737459	$2,845,008,441
Boron Infused Polyethylene	90887091	$6,816,531,825	–	–	–	–
Geopolymer	7706921	$516,363,707	–	–	–	–
Lunar Regolith	10899011	$980,910,990	–	–	–	–
TOTAL	**$32,808,365,144**		**–**		**$24,664,997,018**	
GRAND TOTAL	**$57,473,362,162**					

AUTOMATION COST PER PHASE

	PHASE 1		PHASE 2		PHASE 3	
NAME	**QTY**	**COST**	**QTY**	**COST**	**QTY**	**COST**
ECB	870	$14,752,499,520	465	$9,744,956,640	675	$14,145,904,800
ICB	646	$12,996,178,258	423	$7,663,881,429	806	$14,603,045,938
Neurovision	–	–	–	–	9500	$7,894,186,500
Echo	–	–	–	–	9500	$7,305,224,500
Mindscape	–	–	–		9500	$7,688,036,500
Moonwalker	–	–	–	–	1772	$3,374,632,240
Haloscan	–	–	–	–	1772	$1,782,472,520
Servers	–	–	–	–	17	$1,259,208,000
TOTAL	**$27,748,677,778**		**$17,408,838,069**		**$58,052,710,998**	
GRAND TOTAL	**$103,210,226,845**					

CASSSC LOADS

Water	110
Air	3211
Food	5102
Materials	1,276,732
Seeds	11

EMPLOYEE SALARY COST PER PHASE

	PHASE 1		PHASE 2		PHASE 3	
NAME	**QTY**	**COST**	**QTY**	**COST**	**QTY**	**COST**
Research Lab OM	–	–	–	–	34	$299,144,648
Automation Engineer	40	$48,825,480	91	$111,077,967	54	$65,914,398
Human Safety Engineer	29	$36,999,534	67	$85,481,682	37	$47,206,302
Exterior Construction OM	47	$59,964,762	56	$71,447,376	45	$57,413,070
Interior Construction OM	43	$54,861,378	29	$36,999,534	67	$85,481,682
CTM OM	–	–	23	$154,074,723	34	$227,762,634
TOTAL	**$200,651,154**		**$596,938,386**		**$933,909,086**	
GRAND TOTAL	**$1,731,498,626**					

COST PER PHASE

PHASE 1	
Materials	–
Automation	$27,748,677,778
Employee Salary	$200,651,154
Docks	$160,101,203
Industrial Parks & Research Labs	$181,304,577
Expansion Latch	$249,301,336
TOTAL	**$28,540,036,048**
PHASE 2	
Materials	–
Automation	$17,408,838,069
Employee Salary	$596,938,386
Solar Sail Mechanism	$231,002,559
Space Tugs Interfaces	$110,810,246
Solar Sails	$171,034,701
Communication Modules	$175,600,192
TOTAL	**$18,694,224,153**
PHASE 3	
Materials	–
Automation	$58,052,710,998
Employee Salary	$933,909,086
Utility Routing	$171,009,812
Housing	$155,006,791
Transport Routing	$169,088,209
Recreational Facilities	$89,008,607
Agricultural Facilities	$108,071,992
TOTAL	**$59,678,805,495**
GRAND TOTAL	**$106,913,065,696**